



Cochrane
Library

Cochrane Database of Systematic Reviews

Blue versus white light for transurethral resection of non-muscle invasive bladder cancer (Review)

Maisch P, Hamidi Madani M, Koziarz A, Narayan VM, Kim MH, Dahm P

Maisch P, Hamidi Madani M, Koziarz A, Narayan VM, Kim MH, Dahm P.
Blue versus white light for transurethral resection of non-muscle invasive bladder cancer.
Cochrane Database of Systematic Reviews 2026, Issue 6. Art. No.: CD013776.
DOI: [10.1002/14651858.CD013776.pub3](https://doi.org/10.1002/14651858.CD013776.pub3).

www.cochranelibrary.com

TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	3
SUMMARY OF FINDINGS	4
BACKGROUND	7
OBJECTIVES	8
METHODS	8
RESULTS	12
Figure 1.	14
Figure 2.	16
Figure 3.	17
Figure 4.	18
Figure 5.	19
Figure 6.	19
Figure 7.	21
DISCUSSION	21
AUTHORS' CONCLUSIONS	24
SUPPLEMENTARY MATERIALS	24
ADDITIONAL INFORMATION	24
REFERENCES	26
ADDITIONAL TABLES	34

[Intervention Review]

Blue versus white light for transurethral resection of non-muscle invasive bladder cancer

Philipp Maisch^{1a}, Mohammad Hamidi Madani², Alex Koziarz³, Vikram M Narayan⁴, Myung Ha Kim⁵, Philipp Dahm^{6,7}

¹Department of Urology and Pediatric Urology, University of Ulm, Ulm, Germany. ²Uro-Oncology Research Center, Tehran University of Medical Sciences, Tehran, Iran. ³Diagnostic Radiology, McMaster University, Hamilton, Canada. ⁴Department of Urology, Emory University, Atlanta, Georgia, USA. ⁵Yonsei Wonju Medical Library, Yonsei University Wonju College of Medicine, Wonju, Korea, South. ⁶Urology Section, Minneapolis VA Health Care System, Minneapolis, Minnesota, USA. ⁷Department of Urology, University of Minnesota, Minneapolis, Minnesota, USA

^aORCID iD: <https://orcid.org/0000-0001-8104-167X>

Contact: Philipp Maisch, philippmaisch@gmx.de.

Editorial group: Cochrane Central Editorial Service.

Publication status and date: New search for studies and content updated (conclusions changed), published in Issue 6, 2026.

Citation: Maisch P, Hamidi Madani M, Koziarz A, Narayan VM, Kim MH, Dahm P. Blue versus white light for transurethral resection of non-muscle invasive bladder cancer. *Cochrane Database of Systematic Reviews* 2026, Issue 6. Art. No.: CD013776. DOI: [10.1002/14651858.CD013776.pub3](https://doi.org/10.1002/14651858.CD013776.pub3).

Copyright © 2026 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ABSTRACT

Rationale

Disease recurrence and progression remain major challenges in the treatment of non-muscle invasive bladder cancer (NMIBC). Blue light-enhanced transurethral resection of bladder cancer (TURBT) is an approach to improve staging and achieve a complete resection of NMIBC. Since the first version of this review was published in 2021, new evidence on this topic has become available.

Objectives

To assess the effects of blue light-enhanced TURBT compared to white light-based TURBT in the treatment of non-muscle invasive bladder cancer.

Search methods

We searched several medical literature databases, including the Cochrane Library, MEDLINE, and Embase, as well as two trial registers, with no restrictions on language of publication or publication status, until March 2026.

Eligibility criteria

We included randomized controlled trials using blue light versus white light TURBT. Included participants (older than 18 years) had a high level of suspicion based on imaging or 'visible diagnosis' for primary urothelial carcinoma of the bladder or recurrent urothelial carcinoma of the bladder upon cystoscopy. We excluded studies in which blue light was used in a surveillance setting.

Outcomes

Critical outcomes (primary outcomes) of this review were time to disease recurrence, time to disease progression and serious surgical complications. Important outcomes (secondary outcomes) were time to death from bladder cancer, any adverse events and non-serious surgical complications. The thresholds for clinical relevance of the outcomes were determined through consensus among all authors.

Risk of bias

We assessed the risk of bias using RoB 1. Two review authors independently assessed the risk of bias of each included study. Any disagreements were resolved by consensus or by consultation with a third review author.

Synthesis methods

Two review authors independently performed data extraction and risk of bias assessment. We synthesized results for each outcome using meta-analysis where possible (using a random-effects model with the Mantel-Haenszel method for dichotomous outcomes and the generic inverse-variance method for time-to-event outcomes). Where this was not possible due to the nature of the data, we synthesized results using a narrative approach. We rated the certainty of evidence using the GRADE approach.

Included studies

We included 17 randomized controlled trials involving a total of 4890 participants in the review (1 study with 533 eligible participants was new to this update). The studies compared blue light versus white light TURBT for treatment of NMIBC.

Synthesis of results

Critical outcomes

Blue light TURBT may have little to no effect on the risk of recurrence in people at low risk, but may reduce the risk of recurrence in those at intermediate and high risk (hazard ratio (HR) 0.68, 95% confidence interval (CI) 0.56 to 0.82; low-certainty evidence). For participants with low-, intermediate-, and high-risk NMIBC, this corresponds to 45 (63 fewer to 25 fewer), 102 (145 fewer to 56 fewer), and 137 (200 fewer to 72 fewer) fewer recurrences per 1000 participants when compared to white light TURBT, respectively.

Blue light TURBT probably has little to no effect on the risk of progression in people at low risk, may have little or no effect on the risk of progression in people at intermediate risk, but may reduce the risk of progression in those at high risk (HR 0.70, 95% CI 0.55 to 0.90; low- to moderate-certainty evidence). For participants with low-, intermediate-, and high-risk NMIBC, this corresponds to 1 (1 fewer to 0 fewer), 15 (22 fewer to 5 fewer), and 48 (73 fewer to 16 fewer) fewer progressions per 1000 participants when compared to white light TURBT, respectively.

Blue light TURBT may have little to no effect on serious surgical complications (risk ratio (RR) 0.84, 95% CI 0.30 to 2.29; 2 RCTs, 951 participants; low-certainty evidence). This corresponds to three fewer (13 fewer to 24 more) surgical complications per 1000 participants with blue light TURBT.

Important outcomes

Blue light TURBT may have little to no effect on the risk of death from bladder cancer over time (HR 0.84, 95% CI 0.42 to 1.68; 2 RCTs, 833 participants; low-certainty evidence). This corresponds to 22 deaths per 1000 participants with white light TURBT and three fewer (12 fewer to 14 more) deaths per 1000 participants with blue light TURBT.

Blue light TURBT may have little to no effect on adverse events of any grade (RR 1.07, 95% CI 0.89 to 1.29; 4 RCTs, 1801 participants; low-certainty evidence). This corresponds to 22 more (35 fewer to 92 more) adverse events of any grade per 1000 participants with blue light TURBT.

Blue light TURBT may have little to no effect on non-serious surgical complications (RR 0.90, 95% CI 0.63 to 1.28; 1 RCT; 426 participants; low-certainty evidence). This corresponds to 24 fewer (87 fewer to 66 more) surgical complications per 1000 participants with blue light TURBT.

Authors' conclusions

Blue light-enhanced TURBT for the treatment of non-muscle invasive bladder cancer compared to white light-based TURBT may reduce the risk of disease recurrence and disease progression over time, depending on baseline risk. There may be little or no effect on serious surgical complications. Blue light TURBT may have little to no effect on the time to death from bladder cancer, adverse events or non-serious surgical complications, respectively. The certainty of evidence for our findings was mostly low, meaning that future studies are likely to change the reported estimates of effect. Frequent issues that led us to downgrade the certainty of the evidence were study limitations, inconsistency, and imprecision.

Funding

The update of this review had no dedicated funding.

Registration

Protocol (2020): DOI: 10.1002/14651858.CD013776

Original review (2021): DOI: 10.1002/14651858.CD013776.pub2

PLAIN LANGUAGE SUMMARY

How does a resection (surgical removal) of bladder cancer supported with a special visualization method (blue light) compare to a standard resection with white light in people in whom a tumor of the inner bladder wall is suspected?

Key messages

- Blue light-supported resection (surgical removal) of the bladder may have little to no effect on the chance of cancer returning in people at low risk (risk for cancer returning), but may reduce the chance for those at medium and high risk.
- Blue light-supported resection of the bladder probably has little to no effect on the chance of cancer getting worse in people at low risk (for cancer getting worse) or medium risk, but may reduce the chance for those at high risk.

Why is a special visualization method (blue light) used during resection of the bladder?

In people who might have bladder cancer, the possible tumor is cut from the inner bladder wall using a special instrument inserted through the tube that carries urine from the bladder to outside the body (the urethra) into the bladder. However, it is sometimes difficult to tell what is normal bladder tissue and what is cancer. In order to see the tumor better and remove it completely, a liquid (contrast agent) is put into the bladder through a thin tube (catheter). During surgery, a special light is used to make the cancer cells light up red.

What did we want to find out?

We wanted to find out if using blue light-enhanced resection of the bladder is better than using white light during bladder surgery for people with non-muscle (superficial) invasive bladder cancer. We looked at whether it affects:

- the time it takes for cancer to return (recurrence);
- the time it takes for cancer to get worse (progression);
- whether people lived for longer;
- serious complications during surgery;
- mild complications during surgery;
- and if it had any unwanted effects.

What did we do?

We searched for studies which randomly assigned people with bladder cancer to have either blue light-enhanced resection of the bladder or white light-based resection of the bladder. We compared and summarized the results of these studies and rated our confidence in the evidence, based on factors such as study methods and sizes.

What did we find?

We identified 17 studies comparing the use of blue light-enhanced resection of the bladder to white light-based resection of the bladder. These studies involved a total of 4890 participants in the review. Blue light-enhanced resection of the bladder may have little to no effect on the risk of recurrence in people at low risk, but may reduce the risk of recurrence in those at medium and high risk. Blue light supported resection of the bladder probably has little to no effect on the risk of progression in people at low risk, may have little or no effect on the risk of progression in people at medium risk, but may reduce the risk of progression in those at high risk. We also found that blue light may have little to no effect on the number of mild or serious surgical complications, the risk of death from bladder cancer over time, any unwanted effects and mild surgical complications.

What are the limitations of the evidence?

The most significant limitations of the evidence considered in this review stem from the quality of the included studies. In some studies, the treatments were not delivered exactly as planned, making it harder to compare the groups fairly. Other studies had missing information, and the amount or reasons for missing data differed between treatment groups, which could affect the results. In addition, some studies may have reported only selected outcomes while leaving out others, creating a risk that the findings do not provide a complete picture of the intervention's effects.

How up to date is this evidence?

This review updates our previous review. The evidence is up to date to March 2026.

SUMMARY OF FINDINGS

Summary of findings 1. Blue light compared to white light for transurethral resection of NMIBC

Blue versus white light for transurethral resection of non-muscle invasive bladder cancer

Population: people with non-muscle invasive bladder cancer

Setting: inpatient or outpatient

Intervention: blue light transurethral resection

Comparison of interest: white light transurethral resection

Outcome	Nº of participants (studies)	Certainty of the evidence (GRADE)	Relative effect (95% CI)	Anticipated absolute effects		What happens
				Assumed risk ^a		
				White light	Risk difference with blue light	
Time to disease recurrence (absolute event rates based on 12 months follow-up; MCID 5%)	3420 (16 RCTs)	⊕⊕⊕⊖ LOW ^{c,d} ⊕⊕⊕⊖ LOW ^{c,d} ⊕⊕⊕⊖ LOW ^{c,d}	HR 0.68 (0.56 to 0.82)	Low^b		Blue light TURBT may have little to no effect on the risk of recurrence in people at low risk, but may reduce the risk of recurrence in those at intermediate and high risk.
150 per 1000				45 fewer per 1000 (63 fewer to 25 fewer)		
Intermediate^b						
380 per 1000				102 fewer per 1000 (145 fewer to 56 fewer)		
High^b						
				610 per 1000	137 fewer per 1000 (200 fewer to 72 fewer)	
Time to disease progression (absolute event rates based on 12 months follow-up; MCID 2%)	2626 (10 RCTs)	⊕⊕⊕⊖ MODERATE ^c ⊕⊕⊕⊖ LOW ^{c,e}	HR 0.70 (0.55 to 0.90)	Low^b		Blue light TURBT probably has little to no effect on the risk of progression in people at low risk, may have little or no effect on the risk of progression in people at intermediate risk, but may
				2 per 1000	1 fewer per 1000 (1 fewer to 0 fewer)	

		⊕⊕⊕⊕ LOW ^{c,e}		Intermediate²		reduce the risk of progression in those at high risk.
				50 per 1000	15 fewer per 1000 (22 fewer to 5 fewer)	
				High^b		
				170 per 1000	48 fewer per 1000 (73 fewer to 16 fewer)	
Surgical complications, serious (up to 90 days; MCID 2%)	951 (2 RCTs)	⊕⊕⊕⊕ LOW ^{c,e}	RR 0.84 (0.30 to 2.29)	18 per 1000	3 fewer per 1000 (13 fewer to 24 more)	Blue light TURBT may have little to no effect on serious surgical complications.
Time to death from bladder cancer (absolute event rates based on 60 months follow-up; MCID 2%)	833 (2 RCTs)	⊕⊕⊕⊕ LOW ^{c,f}	HR 0.84 (0.42 to 1.68)	22 per 1000	3 fewer per 1000 (12 fewer to 14 more)	Blue light TURBT may have little to no effect on the time to death from bladder cancer.
Any adverse events (up to 90 days; MCID 5%)	1801 (4 RCTs)	⊕⊕⊕⊕ LOW ^{g,h}	RR 1.07 (0.89 to 1.29)	316 per 1000	22 more per 1000 (35 fewer to 92 more)	Blue light TURBT may have little to no effect on adverse events.
Surgical complications, non-serious (up to 90 days; MCID 5%)	426 (1 RCT)	⊕⊕⊕⊕ LOW ⁱ	RR 0.90 (0.63 to 1.28)	235 per 1000	24 fewer per 1000 (87 fewer to 66 more)	Blue light TURBT may have little to no effect on non-serious surgical complications.

^aWe provide absolute effect size estimates for time to recurrence and time to progression to reflect risk stratification in clinical practice. Corresponding data were not found for time to death from bladder cancer. Baseline risk for other outcomes is assumed to be similar.

^bBaseline risk at 12 months taken from Sylvester and colleagues, based on EORTC risk tables [1].

^cDowngraded by one level for study limitations due to concerns about performance, attrition, and reporting bias.

^dDowngraded by one level for clinically relevant inconsistency ($I^2 > 60\%$).

^eDowngraded by one level for imprecision given that the 95% CI is consistent with both no effect and a clinically important reduction.

^fDowngraded by one level for imprecision given the width of the CI around the relative effect size estimate and uncertainty around the baseline risk.

^gDowngraded by one level for study limitations due to concerns about performance and reporting bias.

^hDowngraded by one level for imprecision given a wide 95% CI, consistent with both no effect and a clinically important increase.

ⁱDowngraded by two levels for imprecision given a wide 95% CI, consistent with both a large increase and large reduction in surgical complications, non-serious.

Abbreviations

CI: confidence interval; EORTC: European Organization for Research and Treatment of Cancer; GRADE: Grading of Recommendations, Assessment, Development and Evaluation; HR: hazard ratio; MCID: minimal clinically important difference; NMIBC: non-muscle invasive bladder cancer; RCT: randomized controlled study; RR: risk ratio; TURBT: transurethral bladder tumor resection

BACKGROUND

Description of the condition

Bladder cancer is the second most common malignancy in urologic cancer patients [2], with a rising incidence in recent decades. In 2022, approximately 614,000 new cases of bladder cancer were reported globally, leading to an estimated 220,000 cancer-related deaths per year, making bladder cancer the ninth most common malignancy worldwide [2].

The most common presenting symptom is hematuria, either macroscopic or microscopic. Other, even less specific, symptoms include recurrent urinary tract infections or irritative voiding symptoms. Some bladder tumors are found incidentally on cross-sectional imaging either in the form of a bladder mass or secondary ureteral obstruction with hydronephrosis, or both.

The diagnostic workup for suspected bladder cancer typically includes a urinalysis and urine culture (to rule out infection), an upper tract assessment such as a computed tomogram with intravenous pyelogram, urine cytology, and a white light office cystoscopy. For both diagnostic and therapeutic purposes, patients then undergo transurethral bladder tumor resection (TURBT), as well as additional bladder biopsies of suspicious-appearing areas as indicated.

At initial diagnosis, about 70% to 75% of patients present with non-muscle invasive tumors [3, 4]. Non-muscle invasive bladder cancer (NMIBC) is defined as tumors that are limited to the mucosa (pTa, carcinoma in situ) or submucosa (pT1) and do not infiltrate the underlying deeper muscle layers [5]. Based on their appearance, NMIBC can either present as a papillary or non-papillary tumor. Involvement of the deep muscle layer of muscularis propria constitutes muscle invasive (pT2) disease or muscle invasive bladder cancer and mandates different, more aggressive management, ideally in the form of radical cystectomy.

NMIBC, especially carcinoma in situ (CIS), is known to have a high risk of tumor recurrence and progression after TURBT. Tumor recurrence at three months and 12 months after TURBT as detected by cystoscopy is reported in up to 30% and 50% of patients, respectively [6]. The five-year recurrence rate is as high as 80%. Progression to muscle invasive bladder cancer may occur in up to 45% of patients within five years [1]. The most frequently used risk classification for NMIBC is based on the European Organization for Research and Treatment of Cancer (EORTC) risk classification, which uses six prognostically relevant pathological and clinical parameters to classify the intravesical recurrence and progression risk into three risk groups: low risk, intermediate risk and high risk [1]. Additional interventions that have been demonstrated to improve the risk of recurrence and potentially progression are various forms of adjuvant intravesical therapy [7, 8].

This is the first update of an existing review published in 2021 [9].

Description of the intervention and how it might work

Effective treatment of NMIBC relies on both accurate staging, including the identification of CIS when present, and the complete resection of all visible tumors. White light cystoscopy of the bladder is the current gold standard procedure for the detection of bladder cancer. However, its sensitivity (6% to 84%) and specificity (43% to 98%) are limited, so not all tumors are always visualized [10]. The

diagnosis of small papillary tumors as well as CIS can be especially difficult, which may result in missed tumors, failure to provide adequate treatment, and cancer progression [10].

Different optical imaging techniques in combination with white light cystoscopy have been investigated to improve the visualization of tumors. Photodynamic diagnosis (PDD), or blue light (synonymous terms), is performed using fluorescent light after intravesical instillation of 5-aminolevulinic acid (5-ALA) or hexaminolevulinic acid (HAL, Hexvix), both for detection only at the time of diagnostic cystoscopy and therapeutically at the time of TURBT. HAL, an ester derivative of 5-ALA with improved pharmacokinetic characteristics, is the only approved drug for blue light in the USA [11]; however, both agents appear comparable in terms of sensitivity and specificity [12].

Blue light is applied by intravesical instillation of the photoactive agents approximately one hour prior to cystoscopy either in the clinic or the operating room. Visualization and TURBT are then performed after drainage of the instilled fluid and activation of a fluorescent light.

Blue light uses photoactive compounds and their interaction with fluorescent light is used to increase the optical difference between normal and malignant tissue [13, 14]. In malignant tissue, a dysregulation in the activity of transport proteins leads to the accumulation of protoporphyrin IX (PPIX) up to 20-fold. PPIX is a precursor of hemoglobin produced during the biosynthesis of 5-ALA, a natural amino acid. Since PPIX is photoactive, a red fluorescence is emitted by excitation at certain wavelengths of light, in particular visible blue light (375 nm to 445 nm) [14, 15]. A limiting factor in the use of 5-ALA is its low depth of penetration into the tissue due to its lipophilic characteristics [13]. In contrast, with HAL, a synthesized ester derivative of 5-ALA, the uptake in cells is increased by passive diffusion of the cell membrane [16]. HAL undergoes local conversion to porphyrins that preferentially accumulate in malignant cells, and the use of blue light with selective filters can highlight these areas within the bladder mucosa (typically with a red appearance before a dark-blue background).

Consequently, with the contrast enhancement of blue light from healthy to malignant tissue, more tumors can be made visible, which can then be resected more completely.

Adverse effects of the intervention

The accumulation of HAL in inflammatory tissue reduces its specificity and may lead to the resection of non-cancerous areas of the bladder, thereby resulting in unnecessary over-treatment of patients [17]. This may potentially result in a higher risk of bleeding, clot retention, and the need for secondary procedures. Other grave complications of more extensive resection include bladder perforation with the need for open surgical repair or tumor spillage into the abdomen, which may result in incurable spread.

In addition, the use of blue light cystoscopy requires changes in workflow, specialized training of staff, and oftentimes capital investment for suitable equipment, which all represent additional costs. Due to this, operative time may be longer.

Why it is important to do this review

Several systematic reviews have investigated the role of blue light in aiding TURBT, and their findings suggest improved cancer

outcomes over white light TURBT. Based on these findings, several widely used evidence-based [18, 19, 20] and consensus-based [21] guidelines support the use of blue light. However, the existing reviews have since become outdated [12, 22, 23], as additional trials have become available and existing trials have provided longer follow-up. In addition, none of the existing systematic reviews have applied the same methodological rigor that is the current standard for a Cochrane review, which includes the comprehensiveness of the search, completion of study screening and data extraction in duplicate, risk of bias assessment on a per-outcome basis, and the use of the GRADE approach to rate the certainty of the evidence [24]. For this reason, we published the first version of this Cochrane review in 2021 [9]. Since further evidence on this topic has become available in the meantime, an update was necessary. The results of this update of the 2021 Cochrane review may therefore provide important new insights to inform guideline recommendations and will add to the existing suite of Cochrane reviews on non-muscle invasive bladder cancer [8, 25, 26, 27, 28].

OBJECTIVES

To assess the effects of blue light-enhanced TURBT compared to white light-based TURBT in the treatment of non-muscle invasive bladder cancer.

METHODS

This review is based on a published protocol [29]. In response to comments arising during external review of the first version of this review, which was published in 2021 [9], we made the following changes.

- We revised our original definition of progression to correspond to that of the International Bladder Cancer Group [30].
- We reconsidered our risk of bias assessment and now consider all outcomes potentially susceptible to detection bias, as their determination includes judgment on the part of the investigators.
- We added a subgroup analysis for the use of 5-aminolevulinic acid (5-ALA) versus hexaminolevulinic acid (HAL), at the request of several external peer referees. This has been clearly labeled as post hoc.

In this updated version of the review, we searched for errata and retractions to verify whether any of the included studies had been withdrawn.

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review [31], and followed PRISMA for the reporting [32].

Criteria for considering studies for this review

Types of studies

We searched for randomized controlled trials (RCTs) to address the comparisons of interest. We planned to exclude pseudo-randomized studies given their greater risk of selection bias, but our search did not identify any. We did not consider cluster-randomized or cross-over trials, as they are not applicable to our review question.

Types of participants

We included participants older than 18 years with a high level of suspicion or 'visible diagnosis' of primary urothelial carcinoma of the bladder or recurrent urothelial carcinoma of the bladder. This high level of suspicion was based on one or more of the following:

- bladder mass or abnormal bladder mucosa findings based on white light office cystoscopy;
- findings suggestive of a bladder mass (bladder filling defect or hydronephrosis, or both) based on cross-sectional imaging;
- positive or atypical urinary cytology (alone or in combination with other markers such as a positive fluorescence in situ hybridization (FISH) test).

We only considered studies of participants without any evidence of distant metastatic disease. We did not consider studies of participants in which blue light was used in a surveillance setting (cystoscopy in a diagnostic setting only).

Types of interventions

We planned to investigate the following comparisons of experimental intervention versus comparator intervention. Concomitant interventions had to be the same in the experimental and comparator groups to establish fair comparisons.

Experimental intervention

The experimental intervention of interest was blue light cystoscopy with 5-ALA or HAL in combination with TURBT.

Comparator intervention

The comparator was white light cystoscopy in combination with TURBT.

Comparison

The review compared blue light cystoscopy via 5-ALA or HAL versus white light cystoscopy.

Outcome measures

The measurement of outcomes assessed in this review was not a study eligibility criterion.

Critical outcomes

- Time to disease recurrence (time-to-event outcome)
- Time to disease progression (time-to-event outcome)
- Surgical complications, serious (grade III, IV, and V according to Clavien-Dindo) (dichotomous outcome) [33]

Important outcomes

- Time to death from bladder cancer (time-to-event outcome)
- Any adverse events (assessed using the National Cancer Institute's Common Terminology Criteria for Adverse Events (CTCAE)) (dichotomous outcome)
- Surgical complications, non-serious (grade I and II according to Clavien-Dindo) (dichotomous outcome)

Method and timing of outcome measurement

- Time to disease recurrence: measured from the time of random sequence generation to the time of any recurrence of bladder cancer (based on TURBT; irrespective of tumor stage or grade)
- Time to disease progression: measured from the time of random sequence generation to the time of progression of bladder cancer as documented by histopathology at the time of TURBT (prompted by abnormal cystoscopy). We defined progression as an increase in tumor stage (defined as lamina propria invasion, e.g. increase from Ta to T1, or CIS to T1; development of muscle invasive disease (stage $\geq$ T2) and development of new lymph node involvement or metastatic disease) or grade (increase in grade from low to high (including CIS) following the definition of the International Bladder Cancer Group) [30].
- Surgical complications, serious: measured within 90 days of initial TURBT according to the Clavien-Dindo classification
- Time to death from bladder cancer: measured from the time of random sequence generation to the time of death from bladder cancer
- Adverse events: adverse events of any grade measured by CTCAE and within 90 days of initial TURBT
- Surgical complications, non-serious: measured within 90 days of initial TURBT according to the Clavien-Dindo classification

Had we been unable to obtain the necessary information to analyze time-to-event outcomes for time to disease recurrence, time to disease progression, and time to death from bladder cancer, we would instead have analyzed these outcomes as dichotomous outcomes up to 12 months (short term) or 13 to 24 months (longer term) after randomization.

We have presented a summary of findings table reporting the following outcomes listed according to priority.

- Time to disease recurrence
- Time to disease progression
- Surgical complications, serious
- Time to death from bladder cancer
- Any adverse events
- Surgical complications, non-serious

Thresholds for clinical relevance of outcomes

- Time to disease recurrence: we assumed the effect to be of clinical relevance if the observed absolute difference was 5% or greater at 12-month follow-up.
- Time to disease progression: we assumed the effect to be of clinical relevance if the observed absolute difference was 2% or greater at 12-month follow-up.
- Surgical complications, serious: we assumed the effect to be of clinical relevance if the observed absolute difference was 2% or greater at initial TURBT or re-resection.
- Time to death from bladder cancer: we assumed the effect to be of clinical relevance if the observed absolute difference was 2% or greater at 12-month follow-up.
- Adverse events of any grade: we assumed the effect to be of clinical relevance if the observed difference was 5% or greater at 12-month follow-up.

- Surgical complications, non-serious: we assumed the effect to be of clinical relevance if the observed difference was 5% or greater at initial TURBT or re-resection.

These thresholds were established based on the input of the clinical experts on the author team considering the relative importance of a given outcome and the expected control event rate.

Search methods for identification of studies

We performed a comprehensive search with no restrictions on language of publication or publication status. We re-ran searches approximately every three to six months for the first version of this review, and on 25 July 2025 and most recently on 31 March 2026 for this update of the review.

Electronic searches

We searched the following databases.

- Cochrane Library (via Wiley; 1970 to 31 March 2026)
 - Cochrane Database of Systematic Reviews (CDSR)
 - Cochrane Central Register of Controlled Trials (CENTRAL; Issue 31 of 3, March 2021)
- MEDLINE (via Ovid; 1946 to 31 March 2026)
- Embase (1947 to 31 March 2026)
- Web of Science Core Collection (1900 to 31 March 2026)
- Scopus (2004 to 31 March 2026)
- LILACS (Latin American and Caribbean Health Sciences Information database; 1982 to 31 March 2026)
- OpenGrey (1997 to 31 March 2026)

We searched the following trial registers.

- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (www.clinicaltrials.gov/; 2000 to 31 March 2026)
- World Health Organization International Clinical Trials Registry Platform (apps.who.int/trialsearch/; 2007 to 31 March 2026)

Our electronic searches also included abstract proceedings of the European Association of Urology (EAU), American Urological Association (AUA), American Society of Clinical Oncology (ASCO), and American Society of Clinical Oncology Genitourinary (ASCO-GU) meetings that are included in the above databases.

Detailed search strategies are shown in [Supplementary material 1](#).

As part of our search, we used the following filters in the respective databases to identify randomized controlled trials.

- MEDLINE: RCT filter - Cochrane Highly Sensitive Search Strategy for identifying randomized trials in MEDLINE: sensitivity-maximizing version (2008 revision); Ovid format (tools.ovid.com/ovidgo/searches/view.php?id=2)
- Embase: RCT filter - Cochrane Highly Sensitive Search Strategy for identifying randomized trials in Embase, amended to embase.com format (sites.google.com/a/york.ac.uk/issg-search-filters-resource/home/rcts/embase-rct-filter)
- Web of Science: RCT filter for Web of Science (WoS) from University of Alberta (ualberta.scholaris.ca/items/4f21a3b7-aba1-4e2a-9bea-8ddcc8cbe4ae)

- Scopus: RCT filter for Scopus from National University of Singapore (libguides.nus.edu.sg/sysreviews/filters)
- LILACS: RCT filter for LILACS from LILACS search tutorial (programa.crics9.org/es/files/2012/08/RoseLILACS_CRICS.pdf)

Searching other resources

We attempted to identify other potentially eligible trials or ancillary publications by searching the reference lists of retrieved included trials, reviews, meta-analyses, and health technology assessment reports. We also contacted study authors of the included trials to identify any further studies that we may have missed. We contacted drug/device manufacturers for ongoing or unpublished trials.

We searched PubMed (pubmed.ncbi.nlm.nih.gov) for errata or retractions of eligible studies within our search on 31 March 2026. On 21 February 2026, we checked the Retraction Watch Database (retractiondatabase.org/RetractionSearch.aspx?) and Retraction Watch (retractionwatch.com) by study title, to check if any of the included studies were retracted, but could not find any retractions.

Data collection and analysis

Selection of studies

We first used the reference management software EndNote to identify and remove duplicate records [34]. We used the reference management software Covidence for the study selection process [35]. Two of three review authors (first version: PM, AK or JV; current version: PM, MHM) independently scanned the titles and abstracts of the remaining records to determine which studies should be assessed further. We investigated all articles deemed potentially relevant as full text. Using Covidence, the same two reviewers categorized the full-text studies as 'included studies' or 'excluded studies.' Any discrepancies were resolved through consensus or through recourse to a third review author (first version: PD; current version: PD). We have presented an adapted PRISMA flowchart documenting the process of study selection and the total number of identified, included, and excluded studies [32]. We listed all articles excluded after full-text screening along with the reasons for their exclusion in the characteristics of excluded studies table in [Supplementary material 3](#).

Data extraction and management

For each included study, two of three review authors (first version: PM, AK or JV; current version: PM, MHM) independently extracted key participant and intervention data using a data extraction form based on guidelines provided in the *Cochrane Handbook for Systematic Reviews of Interventions* [36]. A second review author (first version: AK, JV; current version: MHM) checked these data. Any disagreements were resolved by consensus or by consulting a third review author (first version: PD; current version: PD) if required.

We extracted the following data.

- Study information: author, title, source, publication date, publication type, language, duplicate publications, source of funding, authors' conflicts of interest.
- Study characteristics: study design, randomization method, number of study centers, country of study centers, inclusion and exclusion criteria, subgroup analysis, statistical methods, period of enrollment, follow-up period.

- Participant characteristics: number of participants, number of participants per study arm, age, gender, ethnicity, clinical stage of disease (presentation, focality, tumor size), number of participants recruited/allocated/evaluated.
- Intervention/comparator information: name, dosage, frequency, duration of treatment, adjuvant therapy, re-intervention, follow-up.
- Outcomes: according to our predefined critical and important outcomes (including tumor stage and grade), events of intervention/comparator, timing of outcome measurement, number of re-resections.

We extracted outcome data relevant to this review as needed for calculation of summary statistics and measures of variance. For dichotomous outcomes, we attempted to obtain the numbers of events and totals to populate a 2 x 2 table, as well as summary statistics with corresponding measures of variance. For continuous outcomes, we attempted to obtain means and standard deviations or data necessary to calculate this information. For time-to-event outcomes, we attempted to obtain hazard ratios (HRs) with corresponding measures of variance or data necessary to calculate this information.

Information about potentially relevant ongoing studies, including trial identifiers, is presented in the characteristics of ongoing studies table in [Supplementary material 4](#).

We contacted authors of the included studies if relevant data were missing.

Dealing with duplicate and companion publications

In the event of duplicate publications, companion documents, or multiple reports of a primary study, we maximized the yield of information by mapping all publications to unique studies and collating all available data. We used the most complete data set aggregated across all known publications. In case of doubt, we gave priority to the publication reporting the longest follow-up associated with our critical or important outcomes.

Risk of bias assessment in included studies

Two review authors (first version: PM, AK; current version: PM, MHM) independently assessed the risk of bias of each included study. Any disagreements were resolved by consensus or by consultation with a third review author (first version: PD; current version: PD).

We assessed risk of bias using Cochrane's risk of bias assessment tool RoB 1 [37, 38], which includes the following domains.

- Random sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Other sources of bias

For each study, we judged the risk of bias for each domain as being low, high, or unclear, according to the guidance in the *Cochrane Handbook for Systematic Reviews of Interventions* [37, 38]. We have presented a risk of bias summary figure to illustrate these findings.

For performance bias (blinding of participants and personnel) and detection bias (blinding of outcome assessment), we evaluated the risk of bias separately for each outcome, and grouped outcomes according to whether they were measured subjectively or objectively in the risk of bias tables.

Performance bias

We considered all outcomes similarly susceptible to performance bias.

Detection bias

We considered all outcomes similarly susceptible to detection bias.

Other bias domains

We also assessed attrition bias (incomplete outcome data) on an outcome-specific basis, and presented the judgment for each outcome separately when reporting our findings in the risk of bias tables. We collapsed reporting for outcomes with identical judgments. We considered a < 10% rate of attrition as low; 10% to 20% as unclear; and ≥ 20% in at least one trial arm as a high risk of bias.

We assessed reporting bias on a per-study basis. We classified the risk of bias for this domain as low only if we are able to identify an a priori protocol, and the reporting of outcomes and their analyses matched what the investigators had prespecified.

We further summarized the risk of bias across domains for each outcome in each included study, as well as across studies and domains for each outcome, in accordance with the approach for summary assessments of risk of bias presented in the *Cochrane Handbook for Systematic Reviews of Interventions* [39]. We used the risk of bias assessment on a per-study basis to inform the preplanned sensitivity analyses.

Measures of treatment effect

We expressed dichotomous data as risk ratios (RRs) with 95% confidence intervals (CIs). We expressed time-to-event data as hazard ratios (HRs) with 95% CIs.

Unit of analysis issues

The unit of analysis was the individual participant. Should we identify trials for inclusion with more than two intervention groups in further updates of this review, we will address these in accordance with guidance in the *Cochrane Handbook for Systematic Reviews of Interventions* [40]. For studies with more than two intervention groups (multi-arm studies), we will only consider interventions that address the review objective for a pairwise comparison in our analysis [40]. In the event of repeated reporting of outcome measurements (e.g. at different time points), we used the data representing the longest follow-up.

Dealing with missing data

Where feasible, we attempted to obtain missing data from study investigators. If this information was available, we performed an intention-to-treat (ITT) analysis. If it was not, we performed a modified intention-to-treat (mITT) analysis, adhering to ITT principles with the exception that participants with missing outcome data were excluded. If mITT analysis was not possible, we performed the analysis using the as-treated and per-protocol

population [41]. This was rated as a potential source of bias. We investigated attrition rates (e.g. dropouts, losses to follow-up, and withdrawals) and critically appraised instances of missing data and imputation methods (e.g. last observation carried forward (LOCF)) if used by the study authors. We did not impute missing data for this review.

Reporting bias assessment

In order to identify missing trial data, we searched for completed but not reported trials in the above-mentioned trial registers (see [Electronic searches](#)).

We attempted to obtain study protocols to assess for selective outcome reporting.

If we include 10 or more studies investigating a given outcome in further updates of the review, we will use funnel plots to assess small-study effects. There are several possible explanations for asymmetry of a funnel plot, including true heterogeneity of effect with respect to trial size, poor methodological design (and hence bias of small trials), and publication bias [42]. We will therefore use care in our interpretation of the results.

Synthesis methods

Unless there was good evidence for homogeneous effects across studies, we summarized data using a random-effects model [43]. We interpreted random-effects meta-analyses with due consideration of the whole distribution of effects. In addition, we performed statistical analyses according to the statistical guidelines provided in the *Cochrane Handbook for Systematic Reviews of Interventions* [36]. We used the Mantel-Haenszel method for dichotomous outcomes and the generic inverse-variance method for time-to-event outcomes. We used Review Manager (RevMan) to perform the analyses [44].

In the event of substantial clinical, methodological, or statistical heterogeneity unexplained by subgroup analyses, we did not report outcome results as the pooled effect estimate in a meta-analysis, instead providing a narrative description of the results of each study.

We identified heterogeneity (inconsistency) by visual inspection of the forest plots to assess the amount of overlap of CIs, and by using the I^2 statistic, which quantifies inconsistency across studies to assess the impact of heterogeneity on the meta-analysis [45, 46]. We interpreted the I^2 statistic as follows [47]:

- 0% to 40%: may not be important
- 30% to 60%: may indicate moderate heterogeneity
- 50% to 90%: may indicate substantial heterogeneity
- 75% to 100%: considerable heterogeneity

In the case of heterogeneity, we attempted to determine the potential causes by examining individual study and subgroup characteristics.

Investigation of heterogeneity and subgroup analysis

We expected the following characteristics to introduce clinical heterogeneity, and therefore planned to carry out subgroup analyses with investigation of interactions for each comparison group by:

- setting: primary versus recurrent bladder cancer;
- multifocality: solitary versus multiple lesions of bladder cancer;
- tumor size: tumor size < 3 cm versus ≥ 3 cm; and
- stage: positive cytology and/or history of CIS (in the case of recurrent disease).

We did not include a subgroup analysis for bladder cancer versus urothelial carcinoma of the upper urinary tract as there is an ongoing Cochrane review on this topic [48].

Since we included a new study in the update of this review, we have reconducted all analyses listed in the results section, taking this new study into account.

Our rationale for these subgroup analyses was as follows.

- Setting: according to the European Organization for Research and Treatment of Cancer (EORTC) criteria [1], the setting of primary versus recurrent (primary versus ≤ 1 recurrence versus > 1 recurrence) bladder cancer predicts recurrence and progression; it may represent an effect modifier.
- Multifocality: according to the EORTC criteria [1], the number of tumors (1 versus 2 to 7 versus ≥ 8) predicts recurrence and progression; it may represent an effect modifier.
- Tumor size: according to the EORTC criteria [1], tumor size (< 3 cm versus ≥ 3 cm) predicts recurrence and progression; it may represent an effect modifier.
- Stage: compared to other histological types, the detection of CIS is particularly difficult due to its flat growth within the cell level; the effect of blue light versus white light may vary based on the participant's risk of harboring CIS.

At the request of several external peer referees, we performed one additional post hoc subgroup analysis:

- use of 5-ALA versus HAL as a photodynamic agent in combination with TURBT.

We used the test for subgroup differences in RevMan to compare subgroup analyses if the number of studies was sufficient for our critical outcomes [44].

Equity-related assessment

This updated review is based on a published protocol [29], which did not plan for a formal health equity assessment and therefore, did not systematically capture relevant PROGRESS-Plus characteristics [49, 50], such as place of residence, race/ethnicity/culture/language, occupation, gender, religion, education, socioeconomic status, social capital, and other personal characteristics of included studies. However, we agree that health equity related issues are relevant to this topic and have included such considerations in the discussion.

Sensitivity analysis

We planned to perform sensitivity analyses to explore the influence of the following factors (where applicable) on effect sizes for our critical outcomes.

- Restricting the analyses by taking into account the procedure for re-resection, excluding studies in which all participants underwent re-resection on a routine basis (rather than selectively based on high-risk criteria concordant with current

guidelines such as visible tumor left behind, pT1 high-grade tumors). Routine re-resection may potentially mitigate any benefits of blue light [51, 52, 53].

- Restricting the analyses by taking into account risk of bias, excluding studies at 'high risk' overall.

Certainty of the evidence assessment

We have presented the overall certainty of the evidence for each outcome according to the GRADE approach, which takes into account five criteria not only related to internal validity (risk of bias, inconsistency, imprecision, publication bias), but also to external validity, such as directness of results [54]. For each comparison, two review authors (first version: PM; AK or JV; current version: PM, MHM) independently rated the certainty of evidence for each outcome as 'high,' 'moderate,' 'low,' or 'very low' using GRADEpro GDT [55]. Any discrepancies were resolved through consensus or by arbitration from a third review author (first version: PD; current version: PD) if necessary. For each comparison, we have presented a summary of the evidence for the main outcomes in a summary of findings table, which provides key information regarding the best estimate of the magnitude of the effect in relative terms and absolute differences for each relevant comparison of alternative management strategies; numbers of participants and studies addressing each important outcome; and a rating of the overall confidence in effect estimates for each outcome [56, 57]. If meta-analysis had not been possible, we would have presented the results in a narrative summary of findings table.

We have presented a summary of findings table reporting the following outcomes listed according to a priority rating established by the clinicians on our team with input from external experts.

1. Time to disease recurrence
2. Time to disease progression
3. Surgical complications, serious
4. Time to death from bladder cancer
5. Adverse events
6. Surgical complications, non-serious

We used a minimally contextualized approach that focused on absolute effect size estimates for the interpretation of the results, [58], which we have reported using the proposed language for GRADE narratives [59].

Consumer involvement

Due to limited resources, no consumers were involved in this study. As part of the next update, we will critically evaluate the selected outcomes with the involvement of patient representatives and adjust them if necessary.

RESULTS

Description of studies

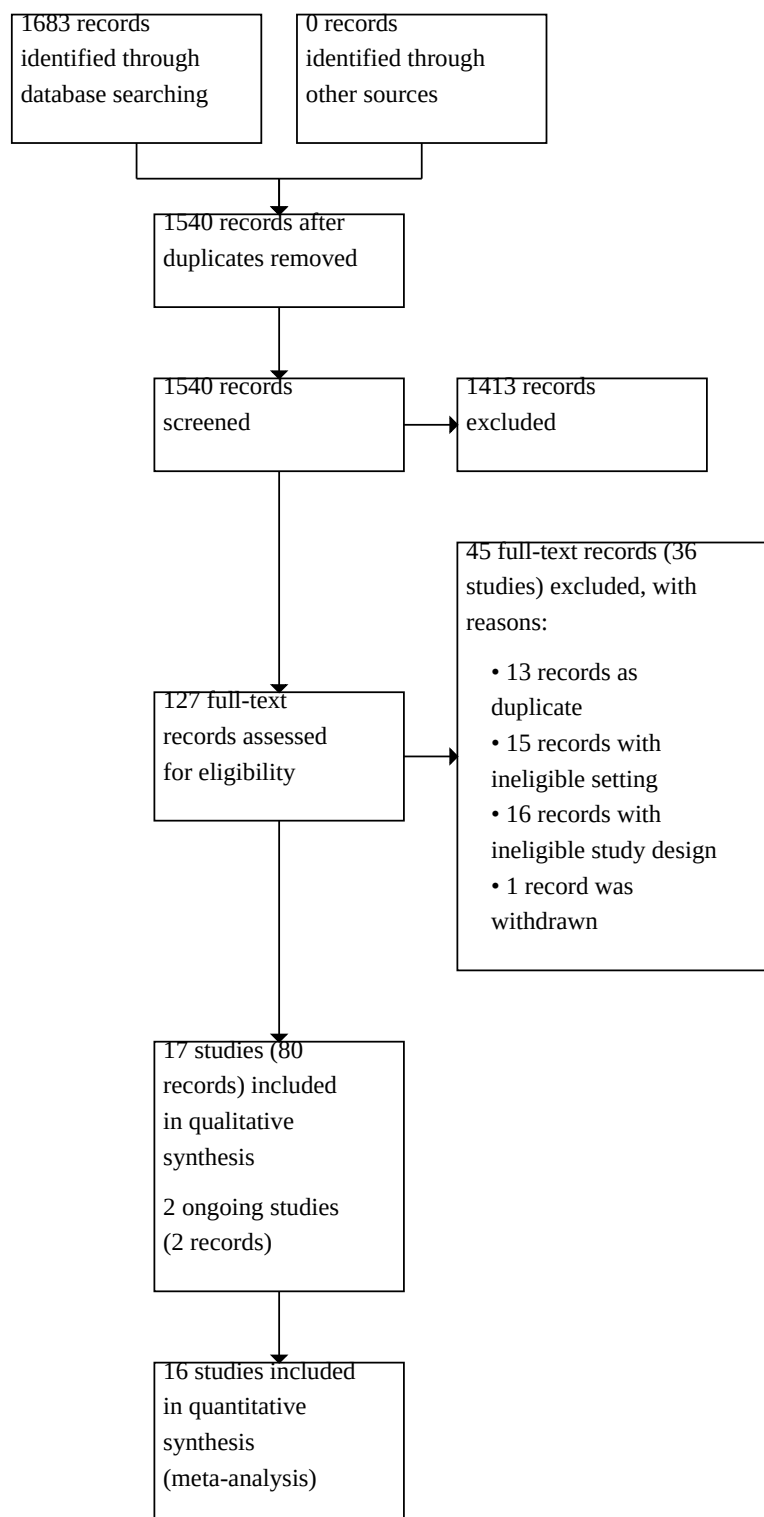
Results of the search

By updating this review, which was first published in 2021, we have identified a total of 1683 references following our database search, of which 143 were excluded as duplicates; we found no records from searching other sources. After title and abstract screening, we excluded 1413 records as irrelevant and retrieved the full texts for 127 records which we assessed for inclusion in the review.

Seventeen studies (80 records) ultimately met the inclusion criteria for assessment of the study question and 45 records were excluded.

The process of study selection is presented in a PRISMA flow diagram (see [Figure 1](#)).

Figure 1. PRISMA study flow diagram



Included studies

We identified 17 trials, including 4890 randomized participants, that were eligible for inclusion (Babjuk 2005 [60, 61]; Drăgoescu 2017 [62, 63, 64, 65, 66]; Filbeck 2002 [67, 68, 69, 70, 71, 72, 73]; Geavlete 2010 [74, 75, 76]; Geavlete 2012 [77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87]; Gkritisios 2014 [88, 89]; Heer 2022 [90, 91, 92, 93, 94, 95, 96]; Hermann 2011 [97, 98]; Karaolides 2012 [99, 100]; Kriegmaier 2002 [101]; Neuzillet 2014 [102]; O'Brien 2013 [103, 104, 105, 106, 107, 108, 109, 110, 111]; Riedl 2001 [112, 113, 114, 115, 116, 117, 118]; Rolevich 2017 [119, 120, 121, 122, 123]; Schumacher 2010 [124, 125]; Stenzl 2010 [126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138]; Stenzl 2011 [139]). The study by Heer 2022, that included 533 eligible participants, was identified within the literature search for this update. From the included studies, we included 3709 participants in the analysis. Of the randomized participants, 2438 received a blue light TURBT and 2452 participants received a white light TURBT.

Full-text publications in the English language were available for all of the included trials. All studies were RCTs; nine were single center, and eight were multicenter. The studies were conducted between 2001 and 2022. Six studies included participants with primary NMIBC, and 11 studies included participants with primary or recurrent NMIBC. Inclusion of participants with suspicion of bladder cancer was based on positive urinary cytology, sonography, cystoscopy, or computed tomography. The trials compared white light TURBT to blue light TURBT using 5-ALA and HAL as the photoactive compound in seven and 10 studies, respectively. Only one study used 50 mL solvent as a comparator in a placebo-controlled setting. Predefined outcomes of the studies were broad and included time to disease recurrence, time to disease progression, time to death from bladder cancer, cancer-specific survival, overall survival, recurrence rate, progression rate, detection rate, residual tumor rate, false-positive rate, surgical complications, and adverse events of any grade. Nine trials reported no funding; four trials reported receiving funding

from pharmaceutical companies; and four trials received funding through national government institutions. Ten studies reported conflict of interest statements, of which five declared they had none.

For the first version of this review, we contacted 15 authors of the included studies for additional information. We received a response from six authors (Babjuk 2005; Hermann 2011; Riedl 2001; Rolevich 2017; Stenzl 2010; Stenzl 2011).

For a detailed description of included studies see [Supplementary material 2](#), [Table 1](#), and [Table 2](#).

Ongoing studies

We identified two ongoing trials addressing our objective (Del Giudice 2025 [140]; NCT06548438 [141]). Recruitment for Del Giudice 2025 is reported to have started on 22 December 2025. At the time of writing, the NCT06548438 trial was not recruiting.

For a detailed description of ongoing studies, see [Supplementary material 4](#).

Excluded studies

During title and abstract screening, we excluded 1413 records that did not meet our inclusion criteria. We excluded 45 records after full-text screening. Reasons for exclusion included ineligible study setting (e.g. surveillance after TURBT; 15 publications); duplicate records (13 publications); ineligible study design (e.g. non-randomized; 16 studies); and study withdrawn before accrual of any participants (1 study; NCT00785694 [142]). For a detailed description of the excluded studies, see [Supplementary material 3](#).

Risk of bias in included studies

For details, see the risk of bias sections in [Supplementary material 2](#), [Figure 2](#), and [Figure 3](#).

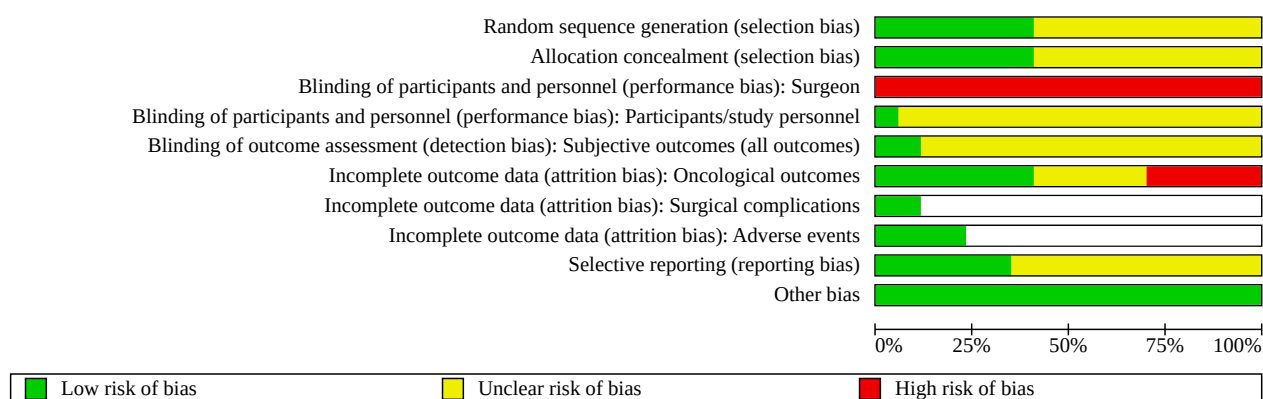
Figure 2. Risk of bias summary: review authors' judgments about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): Surgeon	Blinding of participants and personnel (performance bias): Participants/study personnel	Blinding of outcome assessment (detection bias): Subjective outcomes (all outcomes)	Incomplete outcome data (attrition bias): Oncological outcomes	Incomplete outcome data (attrition bias): Surgical complications	Incomplete outcome data (attrition bias): Adverse events	Selective reporting (reporting bias)	Other bias
Babjuk 2005	?	?	-	?	?	+			?	+
Drăgoescu 2017	?	?	-	?	?	?			?	+
Filbeck 2002	?	?	-	?	?	+			?	+
Geavlete 2010	+	+	-	?	?	?			?	+
Geavlete 2012	+	+	-	?	?	?			?	+
Gkritsios 2014	+	?	-	?	?	-			+	+
Heer 2022	+	+	-	?	+	+	+	+	+	+
Hermann 2011	?	?	-	?	?	-			+	+
Karaolides 2012	?	?	-	?	?	+			?	+
Kriegmaier 2002	?	?	-	?	?	-			?	+
Neuzillet 2014	+	?	-	?	?	+			?	+
O'Brien 2013	+	+	-	?	?	?			+	+
Riedl 2001	?	?	-	?	?	?			?	+

Figure 2. (Continued)

Riedl 2001	?	?	+	?	?	?			?	+
Rolevich 2017	+	+	+	?	?	+	+		?	+
Schumacher 2010	?	+	+	?	?	+		+	?	+
Stenzl 2010	?	+	+	?	?	+		+	+	+
Stenzl 2011	?	?	+	+	+	+		+	+	+

Figure 3. Risk of bias graph: review authors' judgments about each risk of bias item presented as percentages across all included studies

**Random sequence generation**

We assessed the majority of studies (10 of 17) to be at unclear risk of bias for sequence generation (Babjuk 2005; Drăgoescu 2017; Filbeck 2002; Hermann 2011; Karaolides 2012; Kriegmaier 2002; Riedl 2001; Schumacher 2010; Stenzl 2010; Stenzl 2011). We rated the remaining seven studies to be at low risk of bias (Geavlete 2010; Geavlete 2012; Gkritsios 2014; Heer 2022; Neuzillet 2014; O'Brien 2013; Rolevich 2017).

Allocation concealment

Similar to the ratings for random sequence generation, we judged the majority of studies (10 of 17) to be at unclear risk of bias for allocation concealment (Babjuk 2005; Drăgoescu 2017; Filbeck 2002; Gkritsios 2014; Hermann 2011; Karaolides 2012; Kriegmaier 2002; Neuzillet 2014; Riedl 2001; Stenzl 2011). We rated the remaining seven studies to have a low risk of bias (Geavlete 2010; Geavlete 2012; Heer 2022; O'Brien 2013; Rolevich 2017; Schumacher 2010; Stenzl 2010).

Performance bias: blinding of surgeon

Due to the nature of the intervention, the surgeon was not blinded in any of the included studies. Therefore, we rated all studies to have a high risk of bias for this domain.

Performance bias: blinding of participants and other study personnel

Only one study reported blinding of participants and other study personnel; we judged this study to be at low risk of bias (Stenzl 2011). We assessed all other studies as at unclear risk of bias.

Detection bias**Objective outcomes (surgical complications)**

We judged all studies to be at low risk of bias for the objective outcome of surgical complications.

Subjective outcomes (all other outcomes)

Two studies reported blinding of outcome assessors adequately, and we assessed these to be at low risk of bias (Heer 2022; Stenzl 2011). We judged the other studies to be at unclear risk.

Attrition bias

We rated the risk of attrition bias on a per-outcome basis but grouped outcomes with identical ratings together.

Oncological outcomes

We identified seven studies where randomized participants were included adequately in the analysis for oncological outcomes (Babjuk 2005; Filbeck 2002; Heer 2022; Karaolides 2012; Neuzillet 2014; Schumacher 2010; Stenzl 2011). We judged the risk of bias to be unclear for five studies due to attrition rates of 10% to 20% (Drăgoescu 2017; Geavlete 2010; Geavlete 2012; O'Brien 2013; Riedl

2001). We judged five studies to be at high risk of bias for this domain (Gkritisios 2014; Hermann 2011; Kriegmaier 2002; Rolevich 2017; Stenzl 2010).

Surgical complications

Two studies reported the outcome surgical complications (Heer 2022; Rolevich 2017). All participants were included in the analysis; therefore, we rated the studies to be at low risk of bias.

Any adverse events

Only four studies reported this outcome (Heer 2022; Schumacher 2010; Stenzl 2010; Stenzl 2011). In all of these studies, the majority of randomized participants (> 90% per arm) were included in the analysis. Therefore, we judged the risk of attrition bias to be low.

Reporting bias

We failed to identify protocols for the majority of studies (11 of 17); we rated these studies to have an unclear risk of reporting bias (Babjuk 2005; Drăgoescu 2017; Filbeck 2002; Geavlete 2010; Geavlete 2012; Karaolides 2012; Kriegmaier 2002; Neuzillet 2014; Riedl 2001; Rolevich 2017; Schumacher 2010). We identified study protocols for five of the included studies (Gkritisios 2014; Heer 2022; Hermann 2011; O'Brien 2013; Stenzl 2010). One additional study protocol was provided by the trial sponsor (Stenzl 2011). The reported critical and important outcomes corresponded to how they were planned. Therefore, we rated these six studies to be at low risk of reporting bias.

Other bias

We identified no other potential sources of bias.

Synthesis of results

See also [Summary of findings 1](#), [Supplementary material 5](#) and [Supplementary material 6](#).

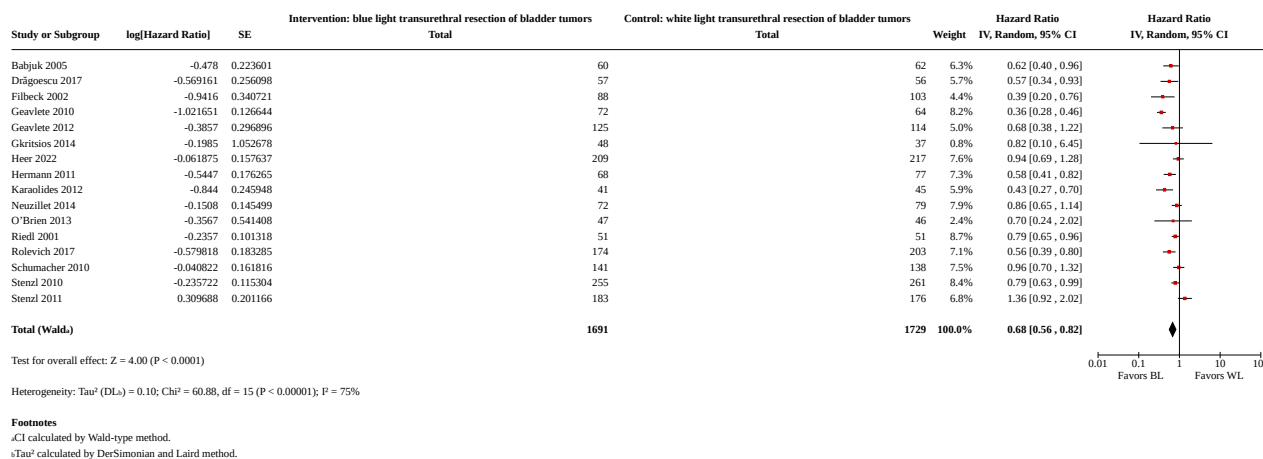
Blue versus white light for transurethral resection of NMIBC

Critical outcomes

Time to disease recurrence

The effect of blue light TURBT on the risk of recurrence was HR 0.68 (95% confidence interval (CI) 0.56 to 0.82; 16 studies, 3420 participants; low-certainty evidence; [Figure 4](#)). Depending on baseline risk, this result indicates that it may have little to no effect on the risk of recurrence in people at low risk, but may reduce the risk of recurrence in those at intermediate and high risk compared to white light TURBT. For participants with low-risk NMIBC with a baseline risk of 15.0% according to EORTC risk categories [1], this corresponds to 45 fewer (63 fewer to 25 fewer) recurrences per 1000, which falls below our predefined threshold for a minimal clinically important difference (MCID) of 50 per 1000. For participants with intermediate- and high-risk NMIBC with baseline risks of 38.0% and 61.0% according to EORTC risk categories [1], this corresponds to 102 (145 fewer to 56 fewer) and 137 (200 fewer to 72 fewer) fewer recurrences per 1000 participants, respectively, when compared to white light TURBT. We downgraded the certainty of the evidence by one level each for study limitations as well as clinically relevant inconsistency.

Figure 4. Forest plot of comparison: blue light versus white light, outcome 1.1. Time to disease recurrence



For three studies, all the information for the analysis was contained in the publications (Heer 2022; Drăgoescu 2017; Rolevich 2017). For nine studies, the HR and P value (if not provided) were calculated with the Parmar method (Geavlete 2010; Geavlete 2012; Gkritisios 2014; Hermann 2011; Karaolides 2012; Neuzillet 2014; O'Brien 2013; Riedl 2001; Stenzl 2011). For the remaining four studies, we performed data reconstruction using digitalization of Kaplan-Meier curves, and calculated HRs and CIs by the Tierney method (Babjuk 2005; Filbeck 2002; Schumacher 2010; Stenzl 2010).

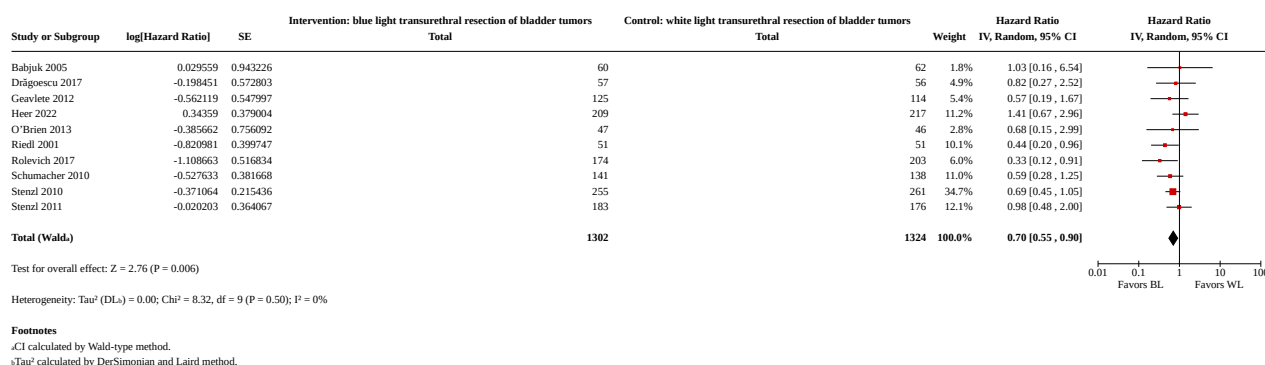
Time to disease progression

The effect of blue light TURBT on the risk of progression was HR 0.70 (95% CI 0.55 to 0.90; 10 studies, 2626 participants; low- (for intermediate- and high-risk NMIBC) to moderate-certainty evidence (for low-grade NMIBC); [Figure 5](#)). Depending on baseline risk, this result indicates that, compared to white light TURBT, blue light TURBT has little to no effect on the risk of progression in people at low risk, may have little or no effect on the risk of progression in people at intermediate risk, but may reduce the risk of progression in those at high risk. For people with low- and intermediate-risk NMIBC with assumed baseline risks of 0.2% and

5% based on EORTC risk categories [1], this corresponds to one fewer (1 fewer to 0 fewer) and 15 fewer (22 fewer to 5 fewer) progressions per 1000 people at 12 months, respectively, when compared to white light TURBT, which falls below our predefined threshold for MCID of 20 per 1000. For people with high-risk NMIBC with an assumed baseline risk of 17.0% per EORTC risk

categories [1], this corresponds to 48 fewer (73 fewer to 16 fewer) progressions per 1000 participants at 12 months. For low-risk NMIBC, we downgraded the certainty of the evidence by one level for study limitations. For intermediate- and high-risk NMIBC, we downgraded the certainty of the evidence by one level each for study limitations as well as imprecision.

Figure 5. Forest plot of comparison: blue light versus white light, outcome 1.2. Time to disease progression



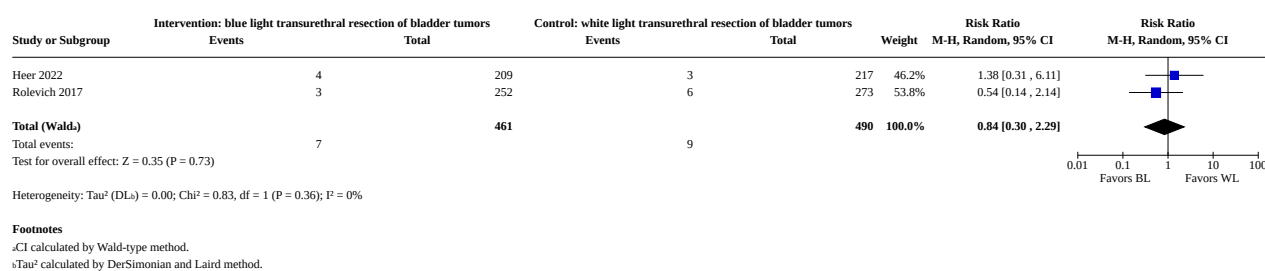
For two studies, all the information for the analysis was contained in the publication (Heer 2022; Rolevich 2017). For six studies, the HR and P value (if not provided) were calculated with the Parmar method (Babjuk 2005; Drăgoescu 2017; Geavlete 2010; O'Brien 2013; Riedl 2001; Stenzl 2010). For two studies, we performed data reconstruction using digitalization of Kaplan-Meier curves, and calculated HRs and CIs with the Tierney method (Schumacher 2010; Stenzl 2011).

Surgical complications, serious

The effect of blue light TURBT on serious surgical complications was RR 0.84 (95% CI 0.30 to 2.29; 2 studies, 951 participants;

low-certainty evidence; Figure 6). Depending on baseline risk, this result indicates that it may have little to no effect on serious surgical complications compared to white light TURBT. Assuming a risk of serious complications of 2.2% in the white light group, this corresponds to three fewer (13 fewer to 24 more) surgical complications per 1000 participants with blue light TURBT, which falls below our predefined threshold for MCID of 20 per 1000. We judged the certainty of evidence to be low (downgraded one level for risk of bias and one level for imprecision).

Figure 6. Forest plot of comparison: blue light versus white light, outcome 1.3. Surgical complications, serious



Important outcomes

Time to death from bladder cancer

The effect of blue light TURBT on the risk of death from bladder cancer over time was HR 0.84 (95% CI 0.42 to 1.68; 2 studies, 833 participants; low-certainty evidence; Analysis 1.4). Depending on baseline risk, this result indicates that it may have little to no effect on the risk of death from bladder cancer over time compared to white light TURBT. Assuming a risk of bladder cancer of 2.2% in the white light group, this corresponds to 22 deaths from bladder cancer per 1000 participants with white light TURBT and three fewer (12 fewer to 14 more) deaths from bladder cancer per 1000 participants with blue light TURBT, which falls below

our predefined threshold for MCID of 20 per 1000. We judged the certainty of evidence to be low (downgraded one level for risk of bias and one level for imprecision).

Any adverse events

The effect of blue light TURBT on adverse events was RR 1.07 (95% CI 0.89 to 1.29; 4 studies, 1801 participants; low-certainty evidence; Analysis 1.5). Depending on baseline risk, this result indicates that it may have little to no effect on adverse events compared to white light TURBT. Assuming a risk of serious complications of 2.2% in the white light group, this corresponds to 22 more (35 fewer to 92 more) adverse events per 1000 participants with blue light TURBT, which falls below our predefined threshold for MCID of 20 per 1000. We

judged the certainty of evidence to be low (downgraded one level for risk of bias and one level for imprecision).

Surgical complications, non-serious

The effect of blue light TURBT on non-serious surgical complications was RR 0.90 (95% CI 0.63 to 1.28; 1 study, 426 participants; low-certainty evidence; Analysis 1.6). Depending on baseline risk, this result indicates that it may have little to no effect on non-serious surgical complications compared to white light TURBT. Assuming a risk of non-serious complications of 23.5% in the white light group, this corresponds to 24 fewer (87 fewer to 66 more) surgical complications per 1000 participants with blue light TURBT. We judged the certainty of evidence to be low (downgraded by two levels for imprecision given that the 95% CI was consistent with both an increase and a reduction in non-serious surgical complications assuming a predefined threshold for MCID of 50 per 1000).

Subgroup analyses

I. Primary versus recurrent bladder cancer

Time to disease recurrence

The pooled effect size for participants with primary bladder tumor (HR 0.79, 95% CI 0.60 to 1.03; 2 studies, 368 participants) was similar to that for participants with recurrent disease (HR 0.80, 95% CI 0.67 to 0.95; 2 studies, 422 participants; Analysis 2.1). All comparisons were across studies and did not suggest a potential subgroup effect ($P = 0.95$).

Time to disease progression

We were unable to perform this analysis due to lack of reported data.

Surgical complications, serious

We were unable to perform this analysis due to lack of reported data.

II. Solitary versus multiple lesions of bladder cancer

Time to disease recurrence

The pooled effect size for participants with a solitary bladder tumor (HR 0.60, 95% CI 0.38 to 0.95; 3 studies, 230 participants) was similar to that for participants with multiple tumors (HR 0.53, 95% CI 0.31 to 0.90; 3 studies, 241 participants; Analysis 3.1). All comparisons were across studies, and we did not find any suggestion of a potential subgroup effect ($P = 0.74$).

Time to disease progression

We were unable to perform this analysis due to lack of reported data.

Surgical complications, serious

No additional analysis was possible as the primary analysis (Analysis 1.3) included only two studies (Heer 2022; Rolevich 2017).

III. Tumor size < 3 cm versus ≥ 3 cm

No analysis was possible as no data were reported by any included study.

IV. Positive cytology and/or history of CIS (in the case of recurrent disease)

No analysis was possible as no data were reported by any included study.

V. 5-ALA versus HAL (post hoc subgroup analysis)

Time to disease recurrence

The pooled effect size of studies using 5-ALA (HR 0.76, 95% CI 0.57 to 1.00; 6 studies, 1430 participants) was similar to that of studies using HAL (HR 0.60, 95% CI 0.45 to 0.78; 9 studies, 1564 participants; Analysis 4.1). All comparisons were across studies, and we did not find any suggestion of a potential subgroup effect ($P = 0.23$).

Time to disease progression

The pooled effect size of studies using 5-ALA (HR 0.72, 95% CI 0.47 to 1.11; 5 studies, 1239 participants) was similar to that of studies using HAL (HR 0.69, 95% CI 0.48 to 0.98; 4 studies, 961 participants; Analysis 4.2). All comparisons were across studies, and we did not find any suggestion of a potential subgroup effect ($P = 0.87$).

Surgical complications, serious

No subgroup analysis was possible as the analysis (Analysis 1.3) included only one study (Rolevich 2017).

Sensitivity analyses

I. Sensitivity analysis by re-resection

We were able to perform sensitivity analyses by excluding studies in which all participants underwent re-resection on a routine basis for the three critical outcomes of this review.

Time to disease recurrence

The effect size (HR 0.64, 95% CI 0.56 to 0.74; 9 studies, 1776 participants; Analysis 5.1) was similar to that of the primary analysis (HR 0.68, 95% CI 0.56 to 0.82; 16 studies, 3420 participants).

Time to disease progression

The effect size (HR 0.64, 95% CI 0.46 to 0.90; 6 studies, 1460 participants; Analysis 5.2) was similar to that of the primary analysis (HR 0.70, 95% CI 0.55 to 0.90; 10 studies, 2626 participants).

Surgical complications, serious

The effect size (HR 1.38, 95% CI 0.31 to 6.11; 1 study, 426 participants; Analysis 5.3) was different to that of the primary analysis (HR 0.84, 95% CI 0.30 to 2.29; 2 studies, 951 participants).

II. Sensitivity analysis by studies at high risk of bias overall

Time to disease recurrence

The effect size (HR 0.94, 95% CI 0.69 to 1.28; 1 study, 426 participants; Analysis 6.1) was different to that of the primary analysis (HR 0.68, 95% CI 0.56 to 0.82; 16 studies, 3420 participants).

Time to disease progression

The effect size (HR 1.41, 95% CI 0.67 to 2.96; 1 study, 426 participants; Analysis 6.2) was different to that of the primary analysis (HR 0.70, 95% CI 0.55 to 0.90; 10 studies, 2626 participants).

Surgical complications, serious

The effect size (HR 1.38, 95% CI 0.31 to 6.11; 1 study, 416 participants; Analysis 6.3) was different to that of the primary analysis (HR 0.84, 95% CI 0.30 to 2.29; 2 studies, 951 participants).

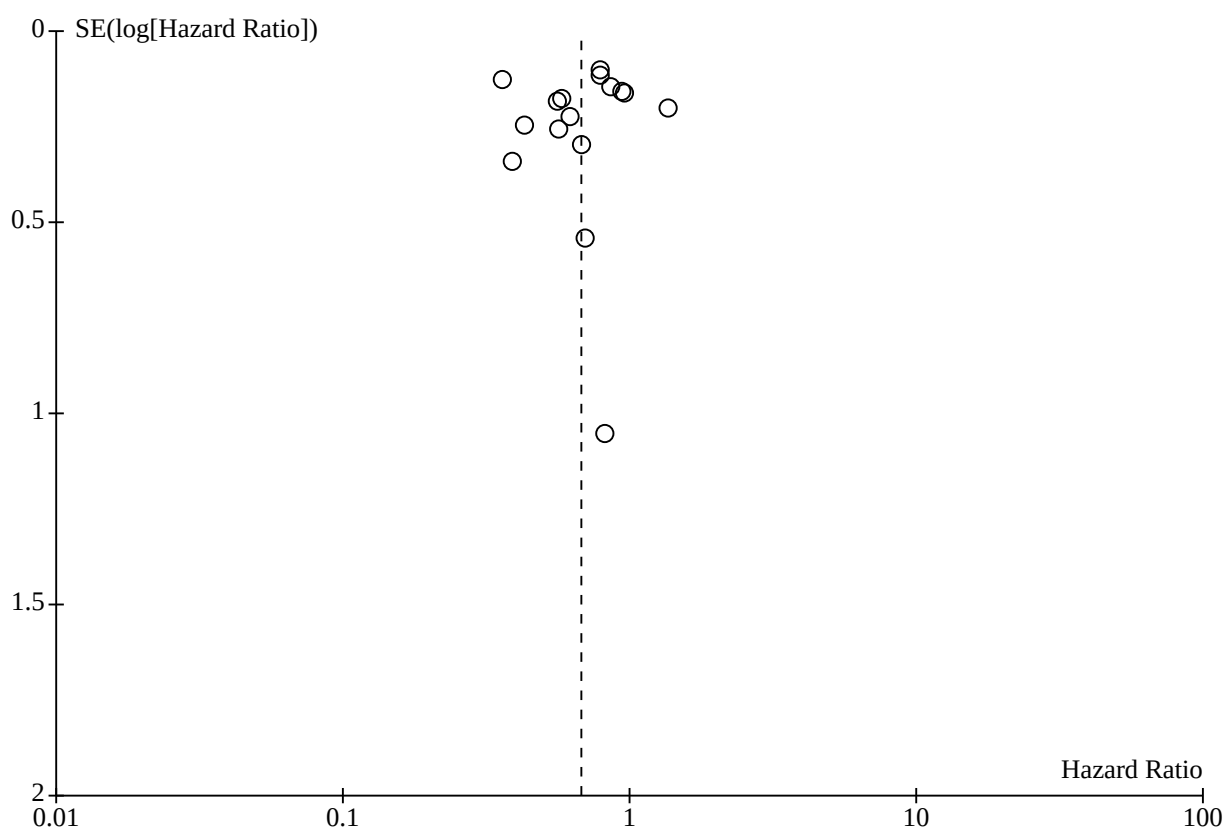
Reporting biases

To minimize reporting bias regarding unpublished studies, we included study registries such as the US National Institutes of Health Ongoing Trials Register (ClinicalTrials.gov) and the World Health Organization International Clinical Trials Registry Platform

(ICTRP) in our systematic literature search. Since registration in study registries was not yet common practice at the time of publication for many of the included studies, we were unable to identify any unpublished studies in our systematic literature search.

Except for one of the included studies, all studies reported the primary outcome of time to disease recurrence. Our analysis does not suggest any reporting bias in respect to this outcome (Figure 7). We did not perform any analysis for other outcomes due to the limited number of included studies.

Figure 7. Funnel plot of comparison: blue light versus white light, outcome 1.1. Time to disease recurrence

**DISCUSSION****Summary of main results**

We identified 17 RCTs, including a total of 4890 participants, that compared blue light-enhanced TURBT with white light-based TURBT in the treatment of primary or recurrent non-muscle invasive bladder cancer. Of 4890 randomized participants, 3709 could be included in the analysis. Of the randomized participants, 2438 received a blue light TURBT and 2452 participants received a white light TURBT.

Blue light TURBT may have little to no effect on the risk of recurrence in people at low risk, but may reduce the risk of recurrence in those at intermediate and high risk (low-certainty evidence). Blue light TURBT probably has little to no effect on the risk of progression in people at low risk, may have little or no effect on the risk of progression in people at intermediate risk, but

may reduce the risk of progression in those at high risk (low- to moderate-certainty evidence). For both outcomes, the magnitude of this treatment effect in absolute terms depends greatly on the baseline risk as reflected by the EORTC prognostic group. Meanwhile, blue light may have little to no effect on the incidence of serious surgical complications (low-certainty evidence).

We also found that blue light may have little to no effect on the risk of death from bladder cancer over time (low-certainty evidence) and may have little to no effect on adverse events of any grade (low-certainty evidence). Blue light may have little to no effect on the incidence of non-serious surgical complications (low-certainty evidence).

Limitations of the evidence included in the review

Except for the outcome time to disease progression in low-risk carcinomas, we rated the certainty of evidence as low for all

outcomes. Reasons for downgrading of the evidence included concerns over study limitations for risk of bias, inconsistency, and imprecision.

- Study limitations: a majority of studies did not provide assurance of allocation concealment (thereby raising concerns over selection bias); very few studies explicitly reported blinding of study participants and personnel or outcome assessors; and more than half of the included studies had substantial attrition of participants. This was in part due to purposeful exclusion of participants at various time points in the course of each study due to a diagnosis of muscle invasive bladder cancer or histopathological findings without malignancy. Different ways of dealing with excluded participants may have distorted the study results. For example, participants with a diagnosis of muscle invasive bladder cancer who subsequently underwent cystectomy may have experienced surgical complications or adverse events that were not accounted for. In addition, few studies had an a priori protocol that was retrievable, which made it difficult to judge the risk of reporting bias. It is notable that far fewer studies reported on serious surgical complications and adverse events than on time to recurrence or progression. All of these issues prompted us to consistently rate the certainty of evidence down by one level.
- Inconsistency: we downgraded the certainty of evidence by one level for the outcome time to disease recurrence in accordance with GRADE guidance considering not only the I^2 value, but also the clinical implications of using a minimally contextualized approach [58].
- Imprecision: we downgraded most of the outcomes by one level for imprecision, as confidence intervals for the pooled effect sizes were consistent with both no effect and clinically important reduction.

This Cochrane review is based on 17 RCTs including participants with primary or recurrent non-muscle invasive bladder cancer (stage Ta, T1, and carcinoma in situ), which represents a population of patients routinely managed in clinical practice.

The studies included in this review were published between 2001 and 2022. During this period, technological progress has led to improvements in the instruments used for visualization and resection. Unfortunately, not all the included studies described the equipment used in detail. Accordingly, the influence of the equipment used on the results of the included studies remains unclear. Examples of this include the recent US Food and Drug Administration-approved use of flexible cystoscopes for blue light visualization (used for surveillance) as well as the type of white light cystoscopy equipment used (i.e. high-definition versus standard-definition camera systems).

Aside from the studies' broadly similar comparisons of blue light to white light TURBT, there were differences in the implementation of the individual studies. Both 5-ALA and HAL were used as the fluorescent agent, and furthermore, only one study used a solvent as a placebo instillation. There were also differences in the postoperative instillation regimen. Administrations of mitomycin C as well as epirubicin or doxorubicin were used. Due to the heterogeneity of the treatments used, we could not consider these data in our analysis, which may have affected the results. Another difference was whether a re-TURBT was performed, and if so, at what point in time. In most of the studies, the time between five

and seven weeks after the initial TURBT was chosen, but re-TURBT was rarely performed again under blue light. Overall, all of these differences reflect clinical heterogeneity that may have impacted our study results.

We were unable to address whether blue light TURBT may obviate the need for routine re-resection [52, 53]. In addition, we did not address the issue of cost consequences or effectiveness, which is relevant given the required capital investment and personnel-intensive workflow alterations [143, 144].

A major source of heterogeneity in the included studies was the postoperative use of intravesical chemotherapy (e.g. postoperative mitomycin instillation) as well as the initiation of intravesical induction therapy. These agents are known to impact outcomes [25, 26, 28], and were applied differently in the included studies, therefore representing a potential source of bias.

Lastly, this review indicates a paucity of published, direct trial-derived evidence on adverse events related to the installation of the photodynamic agent used for blue light cytoscopy. A sponsor-supported study on the issue of safety specific to HLA has been published using pooled trial data [145]. It reported no serious adverse events related to the agent, but did not permit trial-specific data attribution and verification that would have permitted inclusion in the review.

As indicated in the [Methods](#) section, no formal equity impact study was planned. However, equity considerations were raised when the previously published version of this review was used at the guideline developers training workshops of the US GRADE Network (28 February to 1 March 2024 in Denver, CO, USA) and the Canadian Urological Association (CUA) (27 June 2024, in Vancouver, BC, Canada) [9], which (in aggregate) were attended by approximately 80 participants. At both workshops, when discussing the domain of equity in the GRADE evidence-to-decision framework in GRADEpro GDT [55], participants judged that any potential recommendation for the use of blue light would "probably reduce equity", since the use of this technology requires a capital investment for equipment that is available from only one company and because the agent used for intravesical instillation itself is costly and (depending on the jurisdiction) may not be covered by the insurance provider.

Limitations of the review processes

In order to minimize potential bias in the review process, we performed a sensitive search strategy in multiple databases. Two review authors independently performed literature screening, data extraction, and risk of bias assessment. Despite a comprehensive and unrestricted search, we may have missed studies (unpublished, non-English).

Some studies did not provide hazard ratios for the outcomes of time to disease recurrence and time to disease progression. We therefore calculated hazard ratios either with the Parmar method [146], or conducted data reconstruction via digitalization of Kaplan-Meier curves and by the Tierney method [147]. Both methods can only reconstruct hazard ratios according to the information available, which may have resulted in bias.

Not all subgroup analyses contain the recommended number of studies and thus may not be adequately powered to detect differences. The subgroup analysis comparing 5-ALA versus HAL was conducted post hoc and was motivated by the comments

of several peer referees suggesting that HAL may have superior oncological outcomes [148]. Interpreting the results in accordance with Cochrane guidance, we did not find evidence to support a subgroup effect.

Agreements and disagreements with other studies or reviews

Relatively few, higher-quality systematic reviews exist on this topic, which are described as follows.

- Mowatt and colleagues published a systematic review and meta-analysis that addressed both the diagnostic accuracy of photodynamic cytoscopy and its clinical effectiveness compared to white light [12]. Based on a literature search up to 2008, the review identified four RCTs with 709 participants (Babjuk 2005; Filbeck 2002; Kriegmaier 2002; Riedl 2001), and reported a substantially reduced risk of recurrence (RR 0.37, 95% CI 0.20 to 0.69) and greater progression-free survival (RR 1.37, 95% CI 1.18 to 1.59) compared to white light. The authors did not rate the certainty of evidence. The risk of bias assessment also reported unclear allocation concealment and failure to blind participants, personnel, or outcome assessors.
- Rink and colleagues reported a similar systematic review that assessed both diagnostic accuracy and comparative effectiveness and identified a total of 13 randomized trials (date of last search October 2012), but double-counted several studies that had presented both initial results and longer-term follow-up [149]. The authors also provided no meta-analysis of their own.
- Tran and colleagues published an evidence report by the Canadian Agency for Drugs and Technologies with a search date of up to 2016 [150]. Their findings suggested both short-term (based on 10 RCTs) and long-term (based on 12 RCTs) reduced risk of recurrence, but no reduced risk of progression. The results of subgroup analyses for both outcomes favored HAL over 5-ALA, but the tests of interaction were not significant, thereby indicating chance as a potential explanation for these findings. The strength of evidence for the outcomes recurrence and progression was described as low and moderate, respectively.
- Chou and colleagues reported a high-quality systematic review on this topic based on a literature search up to September 2015. The review only included studies published as full text, thereby raising concerns about publication bias [22]. Since publication of the review, one study has published extended follow-up data (Drăgoescu 2017), and results of two additional studies have become available (Heer 2022; Rolevich 2017). In addition, our review adds an analysis based on time-to-event outcomes using hazard ratios for both time to recurrence and time to progression, whereas Chou and colleagues used risk ratios that were analyzed for the short term (less than three months), intermediate term (three months to less than one year), and long term (one year or more). Whereas the interpretation of Chou and colleagues focused on the presence or absence of statistical significance, we applied a minimally contextualized approach to GRADE [58], with predefined thresholds of what we considered the MCID to be for each outcome. Lastly, we were able to provide a more nuanced interpretation of our results by applying the relative effect size measures to different baseline risk groups. As a result, our findings provide more support for the notion that blue light cystoscopy-guided TURBT extends time to progression, but that any clinically important effect may be limited to individuals stratified as high risk.
- Sari and colleagues published a recent systematic review and network meta-analysis, which sought to also assess whether a single intravesical chemotherapy installation added to the therapeutic effectiveness of blue light and narrow band imaging-assisted TURBT compared to white light [151]. The findings were consistent with ours in that blue light cystoscopy-assisted TURBT outperformed white light, and the positive effect was accentuated by intravesical chemotherapy, as one might have expected. Missing from this review, however, was any attempt to rate the certainty of evidence. Also, narrow band imaging-assisted TURBT appeared to outperform blue light.
- Motlagh and colleagues published their systematic review and meta-analysis in 2021, including a systematic search up to September 2021 [152]. The systematic search was limited to only two databases and the review focused on the value of single immediate intravesical chemotherapy in combination with narrow band imaging or blue light. The authors reported the recurrence rate after 12 months for blue light in combination with single immediate intravesical chemotherapy and blue light alone to be significantly lower compared to white light. Those results are in line with the above-mentioned review of Sari and colleagues [151].
- The Canadian Ontario Health Technology Assessment, also published in 2021, was a comprehensive review investigating the effectiveness, safety, and cost-effectiveness of blue light and narrow band imaging as adjuncts to white light TURBT, based on a literature search up to April 2020 conducted in several databases [153]. The authors of this assessment used the Cochrane risk of bias tool to assess the risk of bias of each included study and the GRADE approach to assess the quality of the body of evidence [58]. Due to the authors' predefined exclusion criteria for studies in which the proportion of participants with recurrent tumor was more than 30% of the study sample, this review only included five studies for the comparison of blue light and white light TURBT, whereas our review includes 16 studies for this meta-analysis. In the Ontario Health Technology Assessment, only two studies were included for one time-to-event outcome (time to disease recurrence). According to the above-mentioned systematic review by Chou and colleagues, the assessment used risk ratios for the recurrence and progression rate. Due to the strict inclusion and exclusion criteria and the resulting reduced number of included studies, the conclusions drawn from the results of this review may be limited.
- Veeratterapillay and colleagues also investigated the use of blue light versus white light TURBT in their review published in 2021 [24]. The authors included 12 studies in their review identified following a systematic literature search in three databases performed in April 2020. The GRADE approach was used to assess the certainty of evidence, but the wording that is suggested by GRADE to describe the results was not used [58]. Recurrence rates at 12 and 24 months were reported to be lower in the blue light group.
- Zhao and colleagues published the most recent review on this topic in 2023 and also included 12 studies [23]. Results were based on a systematic literature search conducted in February 2022. The risk of bias for individual studies was assessed using Cochrane's risk of bias tool [37], but the authors did not apply any GRADE approach to assess the quality of the body of

evidence. The results in terms of recurrence rate, progression rate and time to disease recurrence were in line with earlier reviews.

AUTHORS' CONCLUSIONS

Implications for practice

The findings of this review suggest there may be a favorable impact of blue light transurethral bladder tumor resection (TURBT) on the risk of disease recurrence and progression; however, whether this risk reduction is clinically relevant greatly depends on the baseline risk of participants. Blue light TURBT may result in little to no difference in time to death from bladder cancer. The evidence suggests that blue light TURBT does not increase severe surgical complications or non-surgical adverse events.

We expect findings of the review to help further inform evidence-based guidelines by major professional organizations such as the American Urological Association (AUA) and the European Association of Urology (EAU). Current EAU guidelines make a conditional recommendation for the use of “methods to improve tumor visualization”, referring to both blue light cystoscopy and narrow band imaging, but further qualify this by “if available” [19]. The latest AUA guidelines (amended in 2024) state that clinicians “should offer” blue light cystoscopy (moderate recommendation) without further qualifiers to “increase detection and decrease recurrence” [18]. Baseline risk as defined by risk category may be an important criterion to help guide the use of blue light technology in clinical practice.

Implications for research

Although we identified 17 randomized controlled trials addressing the objective of the review, the certainty of the evidence is mostly low. Therefore, future high-quality trials are likely to change our results. Future studies should be conducted with greater methodological rigor with regard to safeguards against selection, performance, and detection biases; have a published a priori protocol; and ensure greater completeness of follow-up. Authors should extend follow-up and plan ahead to address questions with regard to participant subgroups.

Future studies should also take into account recent technological developments such as digital imaging technology or LED (light-emitting diode) illumination. It remains unclear how such innovations in white light technology may alter the assessment of the comparative effectiveness of blue light technology. To identify the most effective treatment option for individuals with non-muscle invasive bladder cancer, head-to-head trials comparing white light, blue light, and other therapeutic options, such as narrow band imaging, should be conducted. Furthermore, surveillance strategies combining blue light and flexible cystoscopy could be examined.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD013776.pub3](https://doi.org/10.1002/14651858.CD013776.pub3).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Analyses

Supplementary material 6 Data package

ADDITIONAL INFORMATION

Acknowledgements

We acknowledge the support of all members of the Cochrane Urology Group.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Dr Andrew Shepherd, Royal Adelaide Hospital, Adelaide, Australia; Adelaide University, Adelaide, Australia;
- Managing Editor (provided editorial guidance to authors, edited the article): Joanne Duffield, Cochrane Editorial Service;
- Editorial Assistant (selected peer reviewers, conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Andrew Savage, Cochrane Editorial Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Muhammad Omer, Allama Iqbal Medical College Lahore (patient and public review); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

Update of this review

- Study selection: PM, MHK, MHM, AK, PD
- Data extraction: PM, MHM
- Entering data into RevMan: PM
- Carrying out the analysis: PM, MHM, PD
- Interpreting the analysis: PM, MHM, VMN, PD
- Disagreement resolution: PD

First version of this review

- Drafting the protocol: PM, PD, VN, AK, JV, MHK
- Study selection: PM, AK, JV, PD
- Data extraction: PM, AK, JV
- Entering data into Review Manager 5: PM
- Carrying out the analysis: PM, PD
- Interpreting the analysis: PM, PD, VMN
- Disagreement resolution: PM

One author involved in the previous version of this review, published in 2021, is no longer included on the author byline: Jon Vajgrt. Some of the content retained in this review reflects his contribution.

Declarations of interest

PM, AK, MHM, VMN, MHK, and PD report no conflicts of interest.

MHM and PD work as urologists and sometimes use this treatment in some patients.

PD is the Co-ordinating Editor of Cochrane Urology and a Sign-off Editor with the Cochrane Central Editorial Service. He was not involved in the editorial process for this review.

Sources of support

Internal sources

- Minneapolis VA Healthcare system, USA
Philipp Dahm received salary support.

External sources

- None, Other
No external sources of support received.

Registration and protocol

Protocol (2020): DOI: 10.1002/14651858.CD013776

What's new

Original review (2021): DOI: 10.1002/14651858.CD013776.pub2

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library ([Supplementary material 6](#)).

- Search strategies ([Supplementary material 1](#))
- Characteristics of included studies ([Supplementary material 2](#))
- Characteristics of excluded studies ([Supplementary material 3](#))
- Characteristics of ongoing studies ([Supplementary material 4](#))
- Analyses ([Supplementary material 5](#))
- Data package ([Supplementary material 6](#))

Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from Excel are available from the authors on reasonable request.

Date	Event	Description
11 June 2026	New search has been performed	The entire review was updated based on an updated literature search performed on 25 July 2025, and rerun on 31 March 2026. The updated literature search identified one new study including 533 eligible participants, and the latest search identified one new ongoing study, which is still not recruiting. As part of the first update, there is a change in authorship. Former author: Jon Vajgrt, new author: Mohammad Hamidi Madani. Changes since the last version of the review and the published protocol can be viewed in the Methods section.
11 June 2026	New citation required and conclusions have changed	As part of the first update, we adapted our conclusions in the updated version of the review. Based on the newly included study, we have adjusted the certainty of evidence for the outcome time to disease progression for low risk non-muscle invasive bladder cancer. Previously, this was 'low' and has now been changed to 'moderate'.

History

Protocol first published: Issue 11, 2020

Review first published: Issue 12, 2021

REFERENCES

1. Sylvester RJ, van der Meijden AP, Oosterlinck W, Witjes JA, Bouffoux C, Denis L, et al. Predicting recurrence and progression in individual patients with stage Ta T1 bladder cancer using EORTC risk tables: a combined analysis of 2596 patients from seven EORTC trials. *European Urology* 2006;**49**(3):466-77. [DOI: [10.1016/j.eururo.2005.12.031](https://doi.org/10.1016/j.eururo.2005.12.031)]
2. Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians* 2024;**74**(2):229-63. [DOI: [10.3322/caac.21834](https://doi.org/10.3322/caac.21834)]
3. Burger M, Catto JW, Dalbagni G, Grossman HB, Herr H, Karakiewicz P, et al. Epidemiology and risk factors of urothelial bladder cancer. *European Urology* 2013;**63**(2):234-41. [DOI: [10.1016/j.eururo.2012.07.033](https://doi.org/10.1016/j.eururo.2012.07.033)]
4. Schned AR, Lenz P, Moore LE, Johnson A, Jones M, Kida M, et al. Analysis of the distribution and temporal trends of grade and stage in urothelial bladder cancer in Northern New England from 1994 to 2004. *ISRN Pathology* 2012;**2012**:283670. [DOI: [10.5402/2012/283670](https://doi.org/10.5402/2012/283670)]
5. Netto GJ, Amin MB, Berney DM, Comp  rat EM, Gill AJ, Hartmann A, et al. The 2022 World Health Organization Classification of Tumors of the Urinary System and Male Genital Organs—Part B: Prostate and Urinary Tract Tumors. *European Urology* 2022;**82**(5):469-82. [DOI: [10.1016/j.eururo.2022.07.002](https://doi.org/10.1016/j.eururo.2022.07.002)]
6. Palou J, Hern  ndez C, Solsona E, Abascal R, Burgu  s JP, Rioja C, et al. Effectiveness of hexaminolevulinate fluorescence cystoscopy for the diagnosis of non-muscle-invasive bladder cancer in daily clinical practice: a Spanish multicentre observational study. *BJU International* 2015;**116**(1):37-43. [DOI: [10.1111/bju.13020](https://doi.org/10.1111/bju.13020)]
7. Jones G, Cleves A, Wilt TJ, Mason M, Kynaston HG, Shelley M. Intravesical gemcitabine for non-muscle invasive bladder cancer. *Cochrane Database of Systematic Reviews* 2012, Issue 1. Art. No: CD009294. [DOI: [10.1002/14651858.CD009294.pub2](https://doi.org/10.1002/14651858.CD009294.pub2)]
8. Schmidt S, Kunath F, Coles B, Draeger DL, Krabbe LM, Dersch R, et al. Intravesical bacillus calmette-gu  rin versus mitomycin C for Ta and T1 bladder cancer. *Cochrane Database of Systematic Reviews* 2020, Issue 1. Art. No: CD011935. [DOI: [10.1002/14651858.CD011935.pub2](https://doi.org/10.1002/14651858.CD011935.pub2)]
9. Maisch P, Koziarz A, Vajgrt J, Narayan V, Kim MH, Dahm P. Blue versus white light for transurethral resection of non-muscle invasive bladder cancer. *Cochrane Database of Systematic Reviews* 2021, Issue 12. Art. No: CD013776. [DOI: [10.1002/14651858.CD013776](https://doi.org/10.1002/14651858.CD013776)]
10. Jocham D, Stepp H, Waidelich R. Photodynamic diagnosis in urology: state-of-the-art. *European Urology* 2008;**53**(6):1138-48. [DOI: [10.1016/j.eururo.2007.11.048](https://doi.org/10.1016/j.eururo.2007.11.048)]
11. FDA. Cysview (hexaminolevulinate hydrochloride), for intravesical solution for bladder instillation only. https://www.accessdata.fda.gov/drugsatfda_docs/label/2010/022555s000lbl.pdf (accessed prior to 8 October 2021).
12. Mowatt G, N'Dow J, Vale L, Nabi G, Boachie C, Cook JA, et al. Photodynamic diagnosis of bladder cancer compared with white light cystoscopy: systematic review and meta-analysis. *International Journal of Technology Assessment in Health Care* 2011;**27**(1):3-10. [DOI: [10.1017/S0266462310001364](https://doi.org/10.1017/S0266462310001364)]
13. Krieg RC, Messmann H, Rauch J, Seeger S, Knuechel R. Metabolic characterization of tumor cell-specific protoporphyrin IX accumulation after exposure to 5-aminolevulinic acid in human colonic cells. *Photochemistry and Photobiology* 2002;**76**(5):518-25. [DOI: [10.1562/0031-8655\(2002\)076<0518:mcotcs>2.0.co;2](https://doi.org/10.1562/0031-8655(2002)076<0518:mcotcs>2.0.co;2)]
14. Witjes JA. Cystoscopy in bladder cancer: the case pro. *European Urology Supplements* 2018;**7**(5):426-9. [DOI: [10.1016/j.eursup.2007.12.007](https://doi.org/10.1016/j.eursup.2007.12.007)]
15. Inoue K. 5-aminolevulinic acid-mediated photodynamic therapy for bladder cancer. *International Journal of Urology* 2017;**24**(2):97-101. [DOI: [10.1111/iju.13291](https://doi.org/10.1111/iju.13291)]
16. Zaak D, Karl A, Stepp H, Tritschler S, Tilki D, Burger M, et al. Fluorescence cystoscopy at bladder cancer: present trials [Die fluoreszenzzytoskopie beim harnblasenkarzinom: aktueller stellenwert]. *Urologe A* 2007;**46**(11):1519-27. [DOI: [10.1007/s00120-007-1563-7](https://doi.org/10.1007/s00120-007-1563-7)]
17. Ray ER, Chatterton K, Khan MS, Chandra A, Thomas K, Dasgupta P, et al. Hexylaminolaevulinate fluorescence cystoscopy in patients previously treated with intravesical bacille Calmette-Gu  rin. *BJU International* 2010;**105**(6):789-94. [DOI: [10.1111/j.1464-410X.2009.08839.x](https://doi.org/10.1111/j.1464-410X.2009.08839.x)]
18. Holzbeierlein JM, Bixler BR, Buckley DI, Chang SS, Holmes R, James AC, et al. Diagnosis and treatment of non-muscle invasive bladder cancer: AUA/SUO Guideline: 2024 Amendment. *Journal of Urology* 2024;**212**(6):936. [DOI: [10.1097/JU.0000000000004251](https://doi.org/10.1097/JU.0000000000004251)]
19. Gontero P, Birtle A, Capoun O, Comp  rat E, Dominguez-Escrig JL, Liedberg F, et al. European Association of Urology Guidelines on Non-muscle-invasive Bladder Cancer (TaT1 and Carcinoma In Situ)-A Summary of the 2024 Guidelines Update. *European Urology* 2024;**86**(6):531-49. [DOI: [10.1016/j.eururo.2024.07.027](https://doi.org/10.1016/j.eururo.2024.07.027)]
20. National Institute for Health and Care Excellence. Bladder cancer: diagnosis and management, NICE guideline [NG2], 2015. Available from <https://www.nice.org.uk/guidance/ng2>.
21. Flaig TW, Spiess PE, Abern M, Agarwal N, Bangs R, Buyyounouski MK, et al. National Comprehensive Cancer Network Guidelines Version 7.2024 Bladder Cancer, 2025. Available from https://www.nccn.org/professionals/physician_gls/pdf/bladder.pdf (accessed prior to 22 February 2025).
22. Chou R, Selph S, Buckley DI, Fu R, Griffin JC, Grusing S, et al. Comparative effectiveness of fluorescent versus white light

cystoscopy for initial diagnosis or surveillance of bladder cancer on clinical outcomes: systematic review and meta-analysis. *Journal of Urology* 2017;**197**(3 Pt 1):548-58. [DOI: [10.1016/j.juro.2016.10.061](https://doi.org/10.1016/j.juro.2016.10.061)]

23. Zhao H, Peng P, Luo Z, Liu H, Sun J, Wang X, et al. Comparison of hexaminolevulinate (HAL)-guided versus white light transurethral resection for NMIBC: a systematic review and meta-analysis of randomized controlled trials. *Photodiagnosis and Photodynamic Therapy* 2023;**41**:103220. [DOI: [10.1016/j.pdpdt.2022.103220](https://doi.org/10.1016/j.pdpdt.2022.103220)]

24. Veeratterapillay R, Gravestock P, Nambiar A, Gupta A, Aboumarzouk O, Rai B, et al. Time to turn on the blue lights: a systematic review and meta-analysis of photodynamic diagnosis for bladder cancer. *European Urology Open Science* 2021;**31**:17-27. [DOI: [10.1016/j.euro.2021.06.011](https://doi.org/10.1016/j.euro.2021.06.011)]

25. Han MA, Maisch P, Jung JH, Hwang JE, Narayan V, Cleves A, et al. Intravesical gemcitabine for non-muscle invasive bladder cancer. *Cochrane Database of Systematic Reviews* 2021, Issue 6. Art. No: CD009294. [DOI: [10.1002/14651858.CD009294.pub3](https://doi.org/10.1002/14651858.CD009294.pub3)]

26. Hwang EC, Sathianathan NJ, Jung JH, Kim MH, Dahm P, Risk MC. Single-dose intravesical chemotherapy after nephroureterectomy for upper tract urothelial carcinoma. *Cochrane Database of Systematic Reviews* 2019, Issue 5. Art. No: CD013160. [DOI: [10.1002/14651858.CD013160.pub2](https://doi.org/10.1002/14651858.CD013160.pub2)]

27. Jung JH, Gudeloglu A, Kiziloz H, Kuntz GM, Miller A, Konety BR, et al. Intravesical electromotive drug administration for non-muscle invasive bladder cancer. *Cochrane Database of Systematic Reviews* 2017, Issue 9. Art. No: CD011864. [DOI: [10.1002/14651858.CD011864.pub2](https://doi.org/10.1002/14651858.CD011864.pub2)]

28. Shepherd AR, Shepherd E, Brook NR. Intravesical Bacillus Calmette-Guérin with interferon-alpha versus intravesical Bacillus Calmette-Guérin for treating non-muscle-invasive bladder cancer. *Cochrane Database of Systematic Reviews* 2017, Issue 3. Art. No: CD012112. [DOI: [10.1002/14651858.CD012112.pub2](https://doi.org/10.1002/14651858.CD012112.pub2)]

29. Maisch P, Koziarz A, Vajgrt J, Narayan V, Kim MH, Dahm P. Blue versus white light for transurethral resection of non-muscle invasive bladder cancer. *Cochrane Database of Systematic Reviews* 2020, Issue 11. Art. No: CD013776. [DOI: [10.1002/14651858.CD013776](https://doi.org/10.1002/14651858.CD013776)]

30. Lamm D, Persad R, Brausi M, Buckley R, Witjes JA, Palou J, et al. Defining progression in nonmuscle invasive bladder cancer: it is time for a new, standard definition. *Journal of Urology* 2014;**191**(1):20-7. [DOI: [10.1016/j.juro.2013.07.102](https://doi.org/10.1016/j.juro.2013.07.102)]

31. Higgins JP, Lasserson T, Thomas J, Flemyng E, Churchill R. Standards for the conduct of new Cochrane Intervention Reviews, and the planning and conduct of updates. In: Higgins JP, Lasserson T, Thomas J, Flemyng E, Churchill R. *Methodological Expectations of Cochrane Intervention Reviews*. Cochrane: London, Version August 2023. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/mecir-manual#anchor-methodological-expectations-of-cochrane-intervention-reviews-mecir>.

32. Liberati A, Altman DG, Tetzlaff J, Mulrow C, Gøtzsche PC, Ioannidis JPA, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *PLOS Medicine* 2009;**6**(7):e1000100. [DOI: [10.1371/journal.pmed.1000100](https://doi.org/10.1371/journal.pmed.1000100)]

33. Clavien PA, Barkun J, de Oliveira ML, Vauthey JN, Dindo D, Schulick RD, et al. The Clavien-Dindo classification of surgical complications: five-year experience. *Annals of Surgery* 2009;**250**(2):187-96. [DOI: [10.1097/SLA.0b013e3181b13ca2](https://doi.org/10.1097/SLA.0b013e3181b13ca2)]

34. EndNote. Version X9. Thomson Reuters, 2019. Available from <https://www.endnote.com>.

35. Covidence. Version accessed 5 July 2021. Melbourne, Australia: Veritas Health Innovation, 2021. Available at <https://www.covidence.org>.

36. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al. *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.0 (updated July 2019). Cochrane, 2019. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.0>.

37. Higgins JP, Altman DG, Gøtzsche PC, Jüni P, Moher D, Oxman AD, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ* 2011;**343**:d5928.

38. Higgins JP, Altman DG, Sterne JA, editor(s). Chapter 8: Assessing risk of bias in included studies. In: Higgins JP, Green S, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0 (updated March 2011). The Cochrane Collaboration, 2011. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v5.1.0>.

39. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al. *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.2 (updated February 2021). Cochrane, 2021. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.2>.

40. Higgins JP, Eldridge S, Li T, editor(s). Chapter 23: Including variants on randomized trials. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.0 (updated July 2019). Cochrane, 2019. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.0>.

41. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized study. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.0 (updated July 2019). Cochrane, 2019. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.0>.

42. Page MJ, Higgins JP, Sterne JA. Chapter 13: Assessing risk of bias due to missing evidence in a meta-analysis (chapter last updated August 2024). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5

(updated August 2024). Cochrane, 2024. Available from <https://cochrane.org/handbook>.

43. Wood L, Egger M, Gluud LL, Schulz KF, Juni P, Altman DG, et al. Empirical evidence of bias in treatment effect estimates in controlled trials with different interventions and outcomes: meta-epidemiological study. *BMJ* 2008;**336**(7644):601-5. [DOI: [10.1136/bmj.39465.451748.AJ](https://doi.org/10.1136/bmj.39465.451748.AJ)]

44. Review Manager (RevMan). Version 10.4.0. The Cochrane Collaboration, 2026. Available at <https://revman.cochrane.org>.

45. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine* 2002;**21**(11):1539-58. [DOI: [10.1002/sim.1186](https://doi.org/10.1002/sim.1186)]

46. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ* 2003;**327**(7414):557-60. [DOI: [10.1136/bmj.327.7414.557](https://doi.org/10.1136/bmj.327.7414.557)]

47. Deeks JJ, Higgins JP, Altman DG, editor(s). Chapter 10: Analysing data and undertaking meta-analyses. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.0 (updated July 2019). Cochrane, 2019. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.0>.

48. Hwang EC, Sathianathan NJ, Jung JH, Kim MH, Narayan V, Hwang JE, et al. Perioperative systemic chemo-immunotherapy for upper urinary tract urothelial carcinoma. *Cochrane Database of Systematic Reviews* 2020, Issue 3. Art. No: CD013567. [DOI: [10.1002/14651858.CD013567](https://doi.org/10.1002/14651858.CD013567)]

49. Karran EL, Cashin AG, Barker T, Boyd MA, Chiarotto A, Dewidar O, et al. Using PROGRESS-plus to identify current approaches to the collection and reporting of equity-relevant data: a scoping review. *Journal of Clinical Epidemiology* 2023;**163**:70-8. [DOI: [10.1016/j.jclinepi.2023.09.017](https://doi.org/10.1016/j.jclinepi.2023.09.017)]

50. O'Neill J, Tabish H, Welch V, Petticrew M, Pottie K, Clarke M, et al. Applying an equity lens to interventions: using PROGRESS ensures consideration of socially stratifying factors to illuminate inequities in health. *Journal Clinical Epidemiology* 2014;**67**(1):56-64. [DOI: [10.1016/j.jclinepi.2013.08.005](https://doi.org/10.1016/j.jclinepi.2013.08.005)]

51. Bogdan G. Editorial comment on: 'Primary complete transurethral resection of bladder tumor using photodynamic diagnosis for high-risk nonmuscle invasive bladder cancer: is a restaging photodynamic transurethral resection really necessary?' by Tadrict et al. *Journal of Endourology* 2021;**35**(7):1047-8.

52. Doisy L, Walz J, Fakhfakh S, Rybikowski S, Koskas Y, Gravis G, et al. Is a routine second transurethral resection of the bladder still necessary after hexaminolevulinate photodynamic diagnosis-assisted TURBT? *Progrès en Urologie* 2019;**29**(6):332-9.

53. Tadrict A, Gondran-Tellier B, McManus R, Al Balushi K, Akiki A, Gaillet S, et al. Primary complete transurethral resection of bladder tumor using photodynamic diagnosis for high-risk nonmuscle invasive bladder cancer: is a restaging

photodynamic transurethral resection really necessary? *Journal of Endourology* 2021;**35**(7):1042-6.

54. Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Schünemann HJ, et al. GRADE: what is "quality of evidence" and why is it important to clinicians? *BMJ* 2008;**336**(7651):995-8. [DOI: [10.1136/bmj.39490.551019.BE](https://doi.org/10.1136/bmj.39490.551019.BE)]

55. GRADEpro GDT. Version accessed 5 July 2021. Hamilton (ON): McMaster University (developed by Evidence Prime), 2021. Available from <https://grade.pro>.

56. Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction—GRADE evidence profiles and summary of findings tables. *Journal of Clinical Epidemiology* 2011;**64**(4):383-94. [DOI: [10.1016/j.jclinepi.2010.04.026](https://doi.org/10.1016/j.jclinepi.2010.04.026)]

57. Schünemann HJ, Higgins JP, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.0 (updated July 2019). Cochrane, 2019. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.0>.

58. Hultcrantz M, Rind D, Akl EA, Treweek S, Mustafa RA, Iorio A, et al. The GRADE Working Group clarifies the construct of certainty of evidence. *Journal of Clinical Epidemiology* 2017;**87**:4-13.

59. Santesso N, Glenton C, Dahm P, Garner P, Akl EA, Alper B, et al. GRADE guidelines 26: Informative statements to communicate the findings of systematic reviews of interventions. *Journal of Clinical Epidemiology* 2020;**119**:126-35.

60. Babjuk M, Soukup V, Petřík R, Jirsa M, Dvořáček J. 5-aminolaevulinic acid-induced fluorescence cystoscopy during transurethral resection reduces the risk of recurrence in stage Ta/T1 bladder cancer. *BJU International* 2005;**96**(6):798-802. [DOI: [10.1111/j.1464-410X.2004.05715](https://doi.org/10.1111/j.1464-410X.2004.05715)]

61. Babjuk M, Soukup V, Petřík R, Pavlík I, Jirsa M, Dvořáček J, et al. Fluorescence cystoscopy in the diagnostics and treatment of superficial urinary bladder tumors [Fluorescencní cystoskopie a její místo v diagnostice a léčbě povrchových nádorů močového mechýře]. *Casopis Lékarů Českých* 2005;**11**(Supl 2):15-8.

62. Drăgoescu PO, Tudorache Ș, Drocaș AI, Mitroi G, Pănuș A, Drăgoescu NAM, et al. Improved diagnosis and long-term recurrence rate reduction for non-muscle-invasive bladder cancer patients undergoing fluorescent hexaminolevulinate photodynamic diagnosis. *Romanian Journal of Morphology and Embryology* 2017;**58**(4):1279-83.

63. Panus A, Dragoescu O, Maria C, Tomescu P. Photodynamic diagnosis and treatment of non-muscle invasive bladder cancer using hexaminolevulinic acid. *Urology* 2013;**82**(3):S211.

64. Dragoescu O, Tomescu P, Panus A, Enache M, Maria C, Stoica L, et al. Photodynamic diagnosis of non-muscle invasive bladder cancer using hexaminolevulinic acid. *Romanian Journal of Morphology and Embryology* 2011;**52**(1):123-7.

- 65.** Dragoescu PO, Enache M, Tomescu PI, Panus A, Maria C, Dragoescu NA, et al. Use of hexaminolevulinate improves diagnosis sensitivity and treatment efficiency of nonmuscle invasive bladder cancer. *European Urology, Supplements* 2011;**10**(9):633.
- 66.** Tomescu PI, Dragoescu PO, Enache MA, Maria C, Panus A, Dragoescu NA, et al. Value of hexaminolevulinate fluorescent cystoscopy and resection in the management of non-muscle invasive bladder cancer. *European Urology, Supplements* 2012;**11**(1):e958.
- 67.** Filbeck T, Pichlmeier U, Knuechel R, Wieland WF, Roessler W. Clinically relevant improvement of recurrence-free survival with 5-aminolevulinic acid induced fluorescence diagnosis in patients with superficial bladder tumors. *Journal of Urology* 2002;**168**(1):67-71.
- 68.** Denzinger S, Wieland WF, Otto W, Filbeck T, Knuechel R, Burger M. Does photodynamic transurethral resection of bladder tumour improve the outcome of initial T1 high-grade bladder cancer? A long-term follow-up of a randomized study. *BJU International* 2008;**101**(5):566-9. [DOI: [10.1111/j.1464-410X.2007.07314.x](https://doi.org/10.1111/j.1464-410X.2007.07314.x)]
- 69.** Filbeck T, Pichlmeier U, Knuechel R, Wieland WF, Roessler W. Clinically relevant reduction of recurrence of superficial bladder tumor with 5-aminolevulinic acid-induced fluorescence diagnosis. Results of a 5 year study [Senkung des Rezidivrisikos oberflächlicher Harnblasenkarzinome mittels 5-Aminolävulinsäure-induzierter Fluoreszenzdiagnostik]. *Urologe A* 2003;**42**(10):1366-73.
- 70.** Otto W, Burger M, Fritsche HM, Blana A, Roessler W, Knuechel R, et al. Photodynamic diagnosis for superficial bladder cancer: do all risk-groups profit equally from oncological and economic long-term results? *Clinical Medicine Insights. Oncology* 2009;**3**:53-8. [DOI: [10.4137/cmo.s1012](https://doi.org/10.4137/cmo.s1012)]
- 71.** Denzinger S, Burger M, Walter B, Knuechel R, Roessler W, Wieland WF, et al. Clinically relevant reduction in risk of recurrence of superficial bladder cancer using 5-aminolevulinic acid-induced fluorescence diagnosis: 8-year results of prospective randomized study. *Urology* 2007;**69**(4):675-9. [DOI: [10.1016/j.urology.2006.12.023](https://doi.org/10.1016/j.urology.2006.12.023)]
- 72.** EUCTR2009-012275-98. Trial to compare the gain of photodynamic diagnostic during transurethral resection of bladder cancer followed by intravesical chemotherapy against white-light-TURB followed by intravesical standard maintenance chemotherapy [Hexvix® versus white light guided turb for EORCT score intermediate risk non-muscle invasive bladder cancer followed by attenuated intravesical adjuvant chemotherapy]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2009-012275-98/DE> (first posted 11 August 2010).
- 73.** Denzinger S, Wieland WF, Otto W, Filbeck T, Knuechel R, Burger M. Does photodynamic transurethral resection of bladder tumour improve the outcome of initial T1 high-grade bladder cancer? A long-term follow-up of a randomized study. *BJU International* 2008;**101**(5):566-9. [DOI: [10.1111/j.1464-410X.2007.07314.x](https://doi.org/10.1111/j.1464-410X.2007.07314.x)]
- 74.** Geavlete B, Jecu M, Multescu R, Georgescu D, Geavlete P. HAL blue-light cystoscopy in high-risk nonmuscle-invasive bladder cancer—re-TURBT recurrence rates in a prospective, randomized study. *Urology* 2010;**76**(3):664-69. [DOI: [10.1016/j.urology.2010.02.067](https://doi.org/10.1016/j.urology.2010.02.067)]
- 75.** Geavlete B, Jecu M, Multescu R, Georgescu D, Geavlete P. HAL blue light cystoscopy in high-risk non-muscle invasive bladder cancer—re-turbt recurrence rates in a prospective, randomized study. *Journal of Endourology* 2010;**24**:A125.
- 76.** Geavlete P, Geavlete B, Jecu M, Multescu R, Georgescu D, Dragutescu M. High-risk non-muscle invasive bladder cancer —“the real deal” for blue light cystoscopy and TURBT? *BJU International* 2010;**106**:6.
- 77.** Geavlete B, Multescu R, Georgescu D, Jecu M, Stanescu F, Geavlete P. Treatment changes and long-term recurrence rates after hexaminolevulinate (HAL) fluorescence cystoscopy: does it really make a difference in patients with non-muscle-invasive bladder cancer (NMIBC)? *BJU International* 2012;**109**(4):549-56. [DOI: [10.1111/j.1464-410X.2011.10374.x](https://doi.org/10.1111/j.1464-410X.2011.10374.x)]
- 78.** Geavlete B, Jecu M, Stanescu F, Moldoveanu C, Multescu R, Georgescu D, et al. Hexaminolevulinate fluorescence versus standard white light cystoscopy—does blue light really make a difference in the long run? *European Urology, Supplements* 2012;**11**(4):88-9.
- 79.** Geavlete B, Multescu R, Georgescu D, Jecu M, Stanescu F, Moldoveanu C, et al. Four-year non-muscle invasive bladder cancer recurrence rates—a prospective, randomized comparison between hexaminolevulinate blue light and standard white light cystoscopy. *Journal of Endourology* 2013;**27**:A147.
- 80.** Geavlete B, Multescu R, Jecu J, Georgescu D, Stanescu F, Geavlete P. Treatment changes and long-term recurrence rates after HAL fluorescence cystoscopy—does blue light really make a difference in the long run? *European Urology, Supplements* 2011;**10**(2):180.
- 81.** Geavlete B, Multescu R, Jecu M, Georgescu D, Stanescu F, Geavlete P. Treatment changes and long-term recurrence rates after HAL fluorescence cystoscopy—does blue light really make a difference in the long run? *Journal of Urology* 2011;**185**(4):e661-2.
- 82.** Geavlete B, Multescu R, Jecu J, Georgescu D, Stanescu F, Geavlete P. Treatment changes and long-term recurrence rates after hal fluorescence cystoscopy: does blue light really make a difference in the long run? *Urology* 2011;**78**(3):S16-7.
- 83.** Geavlete B, Multescu R, Jecu M, Georgescu D, Stanescu F, Moldoveanu C, et al. Treatment changes and long-term recurrence rates after hal fluorescence cystoscopy: a prospective, randomized, long term comparison to the standard approach. *Journal of Endourology* 2011;**25**:A288-9.
- 84.** Geavlete B, Multescu R, Stanescu F, Georgescu D, Jecu M, Moldoveanu C, et al. A prospective, randomized comparison between the hexaminolevulinate blue light and the standard white light cystoscopy concerning the long term recurrence

rates in nonmuscle invasive bladder cancer cases. *European Urology, Supplements* 2012;**11**(1):e955.

85. Geavlete B, Multescu R, Stanescu F, Georgescu D, Jecu M, Moldoveanu C, et al. A prospective, randomized comparison between the hexaminolevulinate blue light and the standard white light cystoscopy concerning the long term recurrence rates in non-muscle invasive bladder cancer cases. *Journal of Urology* 2012;**187**(4):e511-2.

86. Geavlete B, Multescu R, Stanescu F, Moldoveanu C, Jecu M, Georgescu D, et al. Hexaminolevulinate fluorescence versus standard white light cystoscopy: does blue light really make a difference in the long run? *Journal of Endourology* 2012;**26**:A92.

87. Geavlete B, Multescu R, Georgescu D, Jecu M, Stanescu F, Moldoveanu C, et al. Four-year non-muscle invasive bladder cancer recurrence rates—a prospective, randomized comparison between hexaminolevulinate blue light and standard white light cystoscopy. *Journal of Urology* 2013;**189**(4):e523. [DOI: [10.1016/j.juro.2013.02.2633](https://doi.org/10.1016/j.juro.2013.02.2633)]

88. Gkretsios P, Hatzimouratidis K, Kazantzidis S, Dimitriadis G, Ioannidis E, Katsikas V. Hexaminolevulinate-guided transurethral resection of non-muscle-invasive bladder cancer does not reduce the recurrence rates after a 2-year follow-up: a prospective randomized trial. *International Urology and Nephrology* 2014;**46**(5):927-33. [DOI: [10.1007/s11255-013-0603-z](https://doi.org/10.1007/s11255-013-0603-z)]

89. UMIN000008176. Placebo controlled, randomized trial, comparing the efficacy of white light cystoscopy with the fluorescence cystoscopy with the use of hexaminolevulinate, regarding the diagnosis and treatment of bladder cancer. https://center6.umin.ac.jp/cgi-open-bin/ctr_e/ctr_view.cgi?recptno=R000002279 (first posted 15 June 2012).

90. Heer R, Lewis R, Vadiveloo T, Yu G, Mariappan P, Cresswell J, et al. A randomized trial of PHOTodynamic surgery in non-muscle-invasive bladder cancer. *New England Journal of Medicine* 2022;**1**(10):EVIDoA2200092. [DOI: [10.1056/EVIDoA2200092](https://doi.org/10.1056/EVIDoA2200092)]

91. ISRCTN84013636. Photodynamic versus white light-guided treatment of non-muscle invasive bladder cancer. <http://www.isrctn.com/ISRCTN84013636> (first posted 25 July 2014).

92. Tandogdu Z, Lewis R, Duncan A, Penegar S, McDonald A, Vale L, et al. Photodynamic versus white light-guided treatment of non-muscle invasive bladder cancer: a study protocol for a randomised trial of clinical and cost-effectiveness. *BMJ Open* 2019;**99**:e022268.

93. Heer R, Lewis R, Duncan A, Penegar S, Vadiveloo T, Clark E, et al. Photodynamic versus white-light-guided resection of first-diagnosis non-muscle-invasive bladder cancer: PHOTO RCT. *Health Technology Assessment* 2022;**26**(40):1-144. [DOI: [10.3310/PLPU1526](https://doi.org/10.3310/PLPU1526)]

94. Gravestock P, Vadiveloo T, Lewis R, Yu G, Mariappan P, Cresswell J, et al. Can experience of blue light cystoscopy improve a surgeon's white light resection? Data from the

PHOTO trial. *European Urology* 2023;**83**:1004-5. [DOI: [10.1016/S0302-2838\(23\)00755-8](https://doi.org/10.1016/S0302-2838(23)00755-8)]

95. Gravestock P, Vadiveloo T, Lewis R, Duncan A, Penegar S, Clark E, et al. Can experience of blue light cystoscopy improve a surgeon's white light resection? Data from the PHOTO trial. *Journal of Urology* 2023;**209**(Suppl 4):e405. [PMID: 10.1097/JU.0000000000003260.02]

96. Yu G, Rice S, Heer R, Lewis R, Vadiveloo T, Mariappan P, et al. Photodynamic diagnosis-guided transurethral resection of bladder tumour in participants with a first suspected diagnosis of intermediate- or high-risk non-muscle-invasive bladder cancer: cost-effectiveness analysis alongside a randomised controlled trial. *European Urology Open Science* 2023;**53**:67-77. [DOI: [10.1016/j.euros.2023.05.003](https://doi.org/10.1016/j.euros.2023.05.003)]

97. Hermann GG, Mogensen K, Carlsson S, Marcussen N, Duun S. Fluorescence-guided transurethral resection of bladder tumours reduces bladder tumour recurrence due to less residual tumour tissue in Ta/T1 patients—a randomized two-centre study. *BJU International* 2011;**108**(8 Pt 2):E297-303. [DOI: [10.1111/j.1464-410X.2011.10090.x](https://doi.org/10.1111/j.1464-410X.2011.10090.x)]

98. NCT00412971. A randomized, comparative study of Hexvix fluorescence cystoscopy and standard cystoscopy in patients with non-invasive bladder cancer [A randomized, comparative phase III, 2-center study of Hexvix fluorescence cystoscopy and standard cystoscopy in patients with non-invasive bladder cancer]. <https://clinicaltrials.gov/ct2/show/NCT00412971> (first posted 19 December 2006).

99. Karaolides T, Skolarikos A, Bourdounis A, Konandreas A, Mygdalis V, Thanos A, et al. Hexaminolevulinate-induced fluorescence versus white light during transurethral resection of noninvasive bladder tumor: does it reduce recurrences? *Urology* 2012;**80**(2):354-9. [DOI: [10.1016/j.urology.2012.03.067](https://doi.org/10.1016/j.urology.2012.03.067)]

100. Skolarikos A, Karaolides T, Bourdounis A, Logothetis A, Mourmouris P, Konandreas A, et al. Hexaminolevulinate induced fluorescence versus white light during transurethral resection of non-invasive bladder tumor. Does it reduce recurrences? *Journal of Endourology* 2012;**25**:A88.

101. Kriegmair M, Zaak D, Rothenberger KH, Rassweiler J, Jocham D, Eisenberger F, et al. Transurethral resection for bladder cancer using 5-aminolevulinic acid induced fluorescence endoscopy versus white light endoscopy. *Journal of Urology* 2002;**168**(2):475.

102. Neuzillet Y, Methorst C, Schneider M, Lebre T, Rouanne M, Radulescu C, et al. Assessment of diagnostic gain with hexaminolevulinate (HAL) in the setting of newly diagnosed non-muscle-invasive bladder cancer with positive results on urine cytology. *Urologic Oncology* 2014;**32**(8):1135-40. [DOI: [10.1016/j.urolonc.2014.04.005](https://doi.org/10.1016/j.urolonc.2014.04.005)]

103. O'Brien T, Ray E, Chatterton K, Khan MS, Chandra A, Thomas K. Prospective randomized trial of hexylaminolevulinate photodynamic-assisted transurethral resection of bladder tumour (TURBT) plus single-shot intravesical mitomycin C vs conventional white-light TURBT plus mitomycin C in newly presenting non-muscle-invasive

bladder cancer. *BJU International* 2013;**112**(8):1096-104. [DOI: [10.1111/bju.12355](https://doi.org/10.1111/bju.12355)]

104. EUCTR2004-002760-16. A randomised study of "bluelight" hexyl amino levulinic acid (HAL) photodynamic assisted resection of bladder tumours versus conventional "white light" transurethral resection of newly diagnosed bladder tumours comparing rates of tumour recurrence over one year – A phase IV trial. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2004-002760-16/results> (first posted 17 February 2005).

105. Gan C, Chatterton K, Amery S, Ray E, Khan MS, Thomas K, et al. Long term follow up of a prospective randomised trial of Hexylaminolevulinat (HEXVIX®) photodynamic diagnosis (PDD) assisted versus conventional white-light transurethral resection (TURBT) in newly presenting non-muscle invasive bladder cancer (NMIBC). *Journal of Urology* 2014;**191**(4S):e237.

106. Ray E, Chatterton K, Wilby D, Khan MS, Thomas K, O'Brien TS. A prospective randomised trial of Hexylaminolevulinat (HAL) assisted transurethral resection (TURBT) plus single shot intravesical MitomycinC (MMC) versus white light TURBT plus single shot MMC in newly presenting bladder cancer. *BJU International* 2011;**108**:8.

107. Gan C, Chatterton K, Amery S, Ray E, Khan MS, Thomas K, et al. Long term follow up of a prospective randomised trial of Hexylaminolevulinat (HEXVIX®) photodynamic diagnosis (PDD) assisted versus conventional white-light transurethral resection (TURBT) in newly presenting non-muscle invasive bladder cancer (NMIBC). *BJU International* 2014;**113**:8-9.

108. O'Brien TS, Chatterton K, Ray ER, Wilby D, Chandra A, Dickinson F, et al. A prospective randomised trial of Hexylaminolevulinat (Hexvix) assisted transurethral resection (TURBT) plus single shot intravesical mitomycin (MMC) versus conventional white light TURBT plus single shot MMC in newly presenting bladder cancer. *European Urology, Supplements* 2011;**10**(2):150.

109. O'Brien T, Chatterton K, Ray E, Chandra A, Khan S, Thomas K. A prospective randomised trial of hexylaminolevulinat (HAL) assisted transurethral resection (TURBT) plus single shot intravesical mitomycin (MMC) versus white light TURBT plus single shot MMC in newly presenting bladder cancer. *Journal of Urology* 2011;**185**(4):e661.

110. ISRCTN14275387. A randomised study of 'bluelight' hexyl aminolevulinic acid (HAL) photodynamic assisted resection of bladder tumours versus conventional 'whitelight' transurethral resection of newly diagnosed bladder tumours comparing rates of tumour recurrence over one [A randomised study of 'bluelight' hexyl aminolevulinic acid (HAL) photodynamic assisted resection of bladder tumours versus conventional 'whitelight' transurethral resection of newly diagnosed bladder tumours comparing rates of tumour recurrence over one year - a phase IV trial]. <https://www.isrctn.com/ISRCTN14275387> (first posted 2 September 2005).

111. Gan C, Chatterton K, Amery S, Ray E, Khan MS, Thomas K, et al. Long term follow up of a prospective randomised trial of hexylaminolevulinat (Hexvix®) photodynamic diagnosis (PDD) assisted versus conventional white-light transurethral

resection (TURBT) in newly presenting non-muscle invasive bladder cancer (NMIBC). *Journal of Urology* 2014;**191**(4):e237.

112. Riedl CR, Daniltchenko D, Koenig F, Simak R, Loening SA, Pflueger H. Fluorescence endoscopy with 5-aminolevulinic acid reduces early recurrence rate in superficial bladder cancer. *Journal of Urology* 2001;**165**(4):1121-3.

113. Daniltchenko D, Riedl C, Koenig F, Daha LK, Sachs M, Schnorr D. The impact of ALA (5-aminolevulinic acid)-fluorescence detection on the prognosis of superficial bladder cancer. A 1-year survival analysis. *Aktuelle Urologie* 2004;**35**(6):497-501.

114. Danilchenko DI, Koenig F, Riedl K, Schnorr D, Waldman A, Al-Shukri S, et al. Fluorescence-guided transurethral resection of bladder tumours reduces bladder tumour recurrence due to less residual tumour tissue in Ta/T1 patients: a randomized two-centre study. *BJU International* 2011;**108**(8 Pt 2):E297-303.

115. Daniltchenko D, Riedl C, Koenig F, Sachs M, Daha K, Schnorr D, et al. Reducing the risk of cancer recurrence and progression in patients with superficial bladder tumour with 5-aminolevulinic acid-induced fluorescence diagnosis. Results of a prospective randomized 5-year study. *European Urology, Supplements* 2005;**4**(3):221. [DOI: [10.1016/S1569-9056\(05\)80879-6](https://doi.org/10.1016/S1569-9056(05)80879-6)]

116. Sachs MD, Daniltchenko DI, Riedl CR, Koenig F, Daha K, Schnorr D, et al. Fluorescence detection with 5-aminolevulinic acid (ALA) reduces the risk of tumor recurrence and progression in patients with superficial bladder cancer: 5 year results of a prospective randomized trial. *Journal of Urology* 2005;**173**(4 Suppl):246.

117. Sachs MD, Daniltchenko DI, Riedl CR, Koenig F, Daha K, Schnorr D, et al. Fluorescence detection with 5-aminolevulinic acid (ALA) reduces the risk of tumor recurrence and progression in patients with superficial bladder cancer—5 year results of a prospective randomized trial. *Journal of Urology* 2005;**173**(4 Suppl):246.

118. Danilchenko DI, Koenig F, Riedl K, Schnorr D, Waldman A, Al-Shukri S, et al. Clinical significance of macroscopic 5-aminolevulinic acid (ALA)-induced fluorescence in transurethral resection of non-invasive bladder cancer [Klinicheskoe znachenie vizual'noy flouoresentsii indutsirovannoy 5-aminolevulinovoy kislotoy (ALA) pri transuretral'noy rezeksii neinvasivnogo raka mochevogo puzrya]. *Voprosy Onkologii* 2003;**49**(6):734-7.

119. Rolevich AI, Zhegalik AG, Mokhort AA, Minich AA, Vasilevich VY, Polyakov SL, et al. Results of a prospective randomized study assessing the efficacy of fluorescent cystoscopy-assisted transurethral resection and single instillation of doxorubicin in patients with non-muscle-invasive bladder cancer. *World Journal of Urology* 2017;**35**(5):745-52. [DOI: [10.1007/s00345-016-1927-y](https://doi.org/10.1007/s00345-016-1927-y)]

120. Rolevich AI, Sukonko OG, Krasny SA, Polyakov SL, Nabebina TI, Minich AA. Recurrence prevention by combination of photodynamic diagnosis-assisted transurethral resection of the bladder and single postoperative intravesical chemotherapy instillation in patients with nonmuscle-

invasive bladder cancer. *European Urology, Supplements* 2014;**13**(1):e1003.

121. Rolevich AI, Zhegalik AG, Minich AA, Nabebina TI, Polyakov SA, Krasny SA, et al. Transurethral resection guided by photodynamic diagnosis can prevent progression in nonmuscle invasive bladder cancer patients. *European Urology, Supplements* 2016;**15**(3):e208.

122. Rolevich A, Minich A, Vasilevich V, Zhegalik A, Mokhort A, Nabebina T, et al. Efficacy of fluorescent cystoscopy-assisted transurethral resection in patients with non-muscle invasive bladder cancer and quality of surgery: post-hoc analysis of a prospective randomized study. *Central European Journal of Urology* 2019;**72**(4):351-6. [DOI: [10.5173/cej.2019.0003](https://doi.org/10.5173/cej.2019.0003)]

123. Rolevich A, Minich A, Zhegalik A, Mokhort A, Vasilevich V, Nabebina T, et al. Long-term benefit of fluorescent cystoscopy-assisted transurethral resection in patients with non-muscle invasive bladder cancer is dependent on quality of surgery. *European Urology, Supplements* 2017;**16**(5):e2236.

124. Schumacher MC, Holmäng S, Davidsson T, Friedrich B, Pedersen J, Wiklund NP. Transurethral resection of non-muscle-invasive bladder transitional cell cancers with or without 5-aminolevulinic acid under visible and fluorescent light: results of a prospective, randomised, multicentre study. *European Urology* 2010;**57**(2):293-9. [DOI: [10.1016/j.eururo.2009.10.030](https://doi.org/10.1016/j.eururo.2009.10.030)]

125. Schumacher MC, Holmäng S, Davidsson T, Friedrich B, Pedersen J, Wiklund NP. Transurethral resection of non-muscle-invasive bladder transitional cell cancers with or without 5-ALA under visible and fluorescent light-multicenter phase III clinical trial. *Journal of Urology* 2009;**181**(4):688.

126. Stenzl A, Burger M, Fradet Y, Mynderse LA, Soloway MS, Witjes JA, et al. Hexaminolevulinate guided fluorescence cystoscopy reduces recurrence in patients with nonmuscle invasive bladder cancer. *Journal of Urology* 2010;**184**(5):1907-13. [DOI: [10.1016/j.juro.2010.06.148](https://doi.org/10.1016/j.juro.2010.06.148)]

127. Grossman HB, Stenzl A, Fradet Y, Mynderse LA, Kriegmair M, Witjes JA, et al. Long-term decrease in bladder cancer recurrence with hexaminolevulinate enabled fluorescence cystoscopy. *Journal of Urology* 2012;**188**(1):58-62. [DOI: [10.1016/j.juro.2012.03.00](https://doi.org/10.1016/j.juro.2012.03.00)]

128. Burger M, Stenzl A, Mynderse L, Kriegmair M, Witjes JA, Soloway M, et al. Long-term reduction in bladder cancer recurrence with hexaminolevulinate enabled fluorescence cystoscopy. *European Urology, Supplements* 2012;**11**(1):e957.

129. Mynderse L, Stenzl A, Denzinger S, Fradet Y, Soloway MS, Karl A, et al. Hexaminolevulinate fluorescence cystoscopy improves detection and resection of papillary bladder cancer lesions and reduces early recurrences. *Journal of Urology* 2009;**181**(4):689.

130. Grossman HB, Stenzl A, Fradet Y, Mynderse LA, Kriegmair M, Witjes JA, et al. Long-term decrease in bladder cancer recurrence with hexaminolevulinate enabled fluorescence cystoscopy. *Journal of Urology* 2012;**188**(1):58-62.

131. Grossman HB, Stenzl A, Mynderse L, Kriegmair M, Witjes JA, Soloway M, et al. Long-term clinical benefit of hexaminolevulinate enabled fluorescence cystoscopy. *Journal of Urology* 2012;**187**(4):e673-4.

132. Stenzl AS, Roessler WR, Fradet YF, Mynderse LM, Soloway MS, Kriegmair MK, et al. Hexvix® fluorescence cystoscopy improves detection and resection of papillary bladder cancer and reduces early recurrence: a multicentre, prospective, randomized study. *European Urology, Supplements* 2009;**8**(4):373.

133. Grossman HB, Mynderse L, Stenzl A, Burger M, Fradet Y, Soloway M, et al. Hexaminolevulinate new data: results from the recurrence study. *Urology* 2009;**74**(4):S20-1.

134. Stenzl A. Photodynamic diagnosis (PDD) and photodynamic treatment (PRT) in bladder cancer. *International Journal of Urology* 2010;**17**:A106.

135. NCT00233402. Study to improve detection and early recurrence rate in bladder cancer patients using Hexvix fluorescence cystoscopy [A randomized, comparative, controlled phase III, multicenter study of hexvix fluorescence cystoscopy and white light cystoscopy in the detection of papillary bladder cancer and the early recurrence rate in patients with bladder cancer]. <https://clinicaltrials.gov/ct2/show/NCT00233402> (first posted 5 October 2005).

136. Kamat AM, Cookson M, Witjes JA, Stenzl A, Grossman HB. The impact of blue light cystoscopy with hexaminolevulinate (HAL) on progression of bladder cancer—a new analysis. *Bladder Cancer* 2016;**27**(2):273-8. [DOI: [10.3233/BLC-160048](https://doi.org/10.3233/BLC-160048)]

137. DRKS00004103. A randomized, comparative, controlled phase III, multicenter study of Hexvix fluorescence cystoscopy and white light cystoscopy in the detection of papillary bladder cancer and the early recurrence rate in patients with bladder cancer. https://www.drks.de/drks_web/navigate.do?navigationId=trial.HTML&TRIAL_ID=DRKS00004103 (first posted 4 October 2005).

138. EUCTR2004-002259-15. A randomized, comparative, controlled phase III, multicenter study of hexvix fluorescence cystoscopy and white light cystoscopy in the detection of papillary bladder cancer and the early recurrence rate in patients with bladder cancer. <https://www.clinicaltrialsregister.eu/ctr-search/search?query=2004-002259-15> (first posted 20 December 2004).

139. Stenzl A, Penkoff H, Dajc-Sommerer E, Zumbraegel A, Hoeltl L, Scholz M, et al. Detection and clinical outcome of urinary bladder cancer with 5-aminolevulinic acid-induced fluorescence cystoscopy: a multicenter randomized, double-blind, placebo-controlled trial. *Cancer* 2011;**117**(5):938-47. [DOI: [10.1002/cncr.25523](https://doi.org/10.1002/cncr.25523)]

140. Del Giudice F, Vestri A, Fegatelli DA, Hüscher T, Belsey J, Nair R, et al. VI-RADS followed by photodynamic transurethral resection of non-muscle-invasive bladder cancer vs white-light conventional and second-resection: the 'CUT-less' randomised trial protocol. *BJU International* 2025;**135**(2):346-54. [DOI: [10.1111/bju.16531](https://doi.org/10.1111/bju.16531)]

- 141.** NCT06548438. The Impact of PDD During TURB for NMIBC (TUR_BLUE) [The impact of PDD during TURB for NMIBC: a prospective randomized controlled trial]. <https://clinicaltrials.gov/study/NCT06548438> (first posted 12 August 2024).
- 142.** NCT00785694. Recurrence of bladder cancer after transurethral resection with hexvix. clinicaltrials.gov/ct2/show/NCT00785694 (first posted 5 November 2008).
- 143.** Klaassen Z, Li K, Kassouf W, Black PC, Dragomir A, Kulkarni GS. Contemporary cost-consequence analysis of blue light cystoscopy with hexaminolevulinate in non-muscle-invasive bladder cancer. *Canadian Urological Association Journal* 2017;**11**(6):173-81.
- 144.** Witjes JA, Babjuk M, Gontero P, Jacqmin D, Karl A, Kruck S, et al. Clinical and cost effectiveness of hexaminolevulinate-guided blue-light cystoscopy: evidence review and updated expert recommendations. *European Urology* 2014;**66**(5):863-71.
- 145.** Witjes JA, Gomella LG, Stenzl A, Chang SS, Zaak D, Grossman HB. Safety of hexaminolevulinate for blue light cystoscopy in bladder cancer. A combined analysis of the trials used for registration and postmarketing data. *Urology* 2014;**84**(1):122-6.
- 146.** Parmar MK, Torri V, Stewart L. Extracting summary statistics to perform meta-analyses of the published literature for survival endpoints. *Statistics in Medicine* 1998;**17**(24):2815-34. [DOI: [10.1002/\(sici\)1097-0258\(19981230\)](https://doi.org/10.1002/(sici)1097-0258(19981230)17(24)<2815::aid-sim1230>3.0.co;2-1)]
- 147.** Tierney JF, Stewart LA, Ghersi D, Burdett S, Sydes MR. Practical methods for incorporating summary time-to-event data into meta-analysis. *Trials* 2007;**8**(16):1-16. [DOI: [10.1186/1745-6215-8-1](https://doi.org/10.1186/1745-6215-8-1)]
- 148.** Gakis G, Ngamsri T, Rausch S, Mischinger J, Todenhofer T, Schwentner C, et al. Fluorescence-guided bladder tumour resection: impact on survival after radical cystectomy. *World Journal of Urology* 2015;**33**(10):1429-37.
- 149.** Rink M, Babjuk M, Catto JW, Jichlinski P, Shariat SF, Stenzl A, et al. Hexyl aminolevulinate-guided fluorescence cystoscopy in the diagnosis and follow-up of patients with non-muscle-invasive bladder cancer: a critical review of the current literature. *European Urology* 2013;**64**(4):624-38.
- 150.** Tran K, Severn M. Blue light cystoscopy in patients with suspected non-muscle invasive bladder carcinoma: a review of clinical utility; 15 February 2017. Available at https://www.ncbi.nlm.nih.gov/books/NBK447631/pdf/Bookshelf_NBK447631.pdf.
- 151.** Sari Motlagh R, Mori K, Laukhtina E, Aydh A, Katayama S, Grossmann NC, et al. Impact of enhanced optical techniques at time of transurethral resection of bladder tumour, with or without single immediate intravesical chemotherapy, on recurrence rate of non-muscle-invasive bladder cancer: a systematic review and network meta-analysis of randomized trials. *BJU International* 2021;**128**(3):280-9.
- 152.** Motlagh SR, Mori K, Laukhtina E, Aydh A, Katayama S, Grossmann NC, et al. Impact of enhanced optical techniques at time of transurethral resection of bladder tumour, with or without single immediate intravesical chemotherapy, on recurrence rate of non-muscle-invasive bladder cancer: a systematic review and network meta-analysis of randomized trials. *BJU International* 2021;**128**(3):280-9. [DOI: [10.1111/bju.15383](https://doi.org/10.1111/bju.15383)]
- 153.** Ontario Health (Quality). Enhanced visualization methods for first transurethral resection of bladder tumour in suspected non-muscle-invasive bladder cancer: a health technology assessment. Ontario Health Technology Assessment Series 2021;**21**(12):1-123.

ADDITIONAL TABLES

Table 1. Overview of included studies and synthesis

Study name	Trial period (year to year)	Set-ting/country	Description of partici-pants	Intervention(s) and comparator(s)	Adjuvant in-stillation ^a /re-resection	Duration of follow-up (median)	Age (years (mean), stan-dard devia-tion or range)	Gender (male/fe-male)
Babjuk 2005	2001 to 2005	Single cen-ter/Czech Re-public	Primary and recurrent NMIBC, only Ta/T1	Intervention: WLC + BLC + BL-TURBT (1 g 5-ALA)	NA/NA	24 months	67.9 ± NA	43/17
				Comparator: WLC + WL-TURBT			67.9 ± NA	39/23
Drăgoescu 2017	2009 to 2011	Single cen-ter/Romania	Primary NMIBC	Intervention: WLC + BLC + BL-TURBT (0.85 g HAL)	Mitomycin C, doxorubicin, far-morubicin/NA	60 months	59.4 ± 9.9	45/12
				Comparator: WLC + WL-TURBT			60.3 ± 10.2	43/13
Filbeck 2002	1997 to 2000	Single cen-ter/Germany	Primary and recurrent NMIBC	Intervention: BL-TURBT (1 g 5-ALA)	NA/WL-TURBT after 6 weeks	24 months	68.0 (31 to 88)	NA
				Comparator: WL-TURBT			70.0 (32 to 89)	NA
Geavlete 2010	2007 to 2009	Single cen-ter/Romania	Primary NMIBC	Intervention: WLC + BLC + WL-TURBT + BL-TURBT (HAL)	Mitomycin C/ WL-TURBT af-ter 6 weeks	6 weeks	64.0 (32 to 86)	327/119
				Comparator: WLC + WL-TURBT				
Geavlete 2012	NA	Single cen-ter/Romania	Primary and recurrent NMIBC	Intervention: WLC + BLC + WL-TURBT + BL-TURBT (HAL)	Mitomycin C/ NA	24 months	66.8 (31 to 85)	267/95
				Comparator: WLC + WL-TURBT				
Gkritisios 2014	NA	Single cen-ter/Greece	Primary and recurrent NMIBC	Intervention: WLC + BLC + WL-TURBT + BL-TURBT (0.85 g 5-ALA)	Epirubicin/NA	40 months	66.0 ± NA	43/11
				Comparator: WLC + WL-TURBT			68.2 ± NA	44/6
Heer 2022	2014 to 2018	Multicen-ter/United Kingdom	Primary NMIBC	Intervention: BL-TURBT (0.85 g HAL)	Mitomycin C/ WL-TURBT	44 months	71.0 ± NA	NA
				Comparator: WL-TURBT			70.0 ± NA	NA
Hermann 2011	NA	Multicen-ter/Denmark	Primary and recurrent NMIBC, only Ta/T1	Intervention: WLC + WL-TURBT + BL-TURBT (0.85 g 5-ALA)	NA/NA	12 months	71.0 (35 to 96)	NA

Table 1. Overview of included studies and synthesis (Continued)

				Comparator: WLC + WL-TURBT			69.0 (41 to 92)	NA
Karaolides 2012	2008 to 2010	Single center/Greece	Primary and recurrent NMIBC	Intervention: WLC + BLC + WL-TURBT + BL-TURBT (HAL)	Epirubicin/NA	18 months	66.3 (37 to 82)	33/8
				Comparator: WL-TURBT			63.8 (39 to 88)	40/5
Kriegmaier 2002	1997 to 1998	Multicenter/Germany, Austria	Primary and recurrent NMIBC	Intervention: 1 g 5-ALA + BL-TURBT	NA/WL-TURBT after 10 to 14 days	2 weeks	69.3 (38 to 88)	53/12
				Comparator: WL-TURBT			69.6 (34 to 94)	45/19
Neuzillet 2014	2009 to 2012	Multicenter/France	Primary NMIBC	Intervention: WLC + WL-TURBT + BLC + BL-TURBT (0.85 g HAL)	NA/PDD-TURBT after 6 weeks	6 weeks	74.0 ± 10.3	64/8
				Comparator: WLC + WL-TURBT			74.0 ± 10.4	69/10
O'Brien 2013	2005 to 2010	Single center/United Kingdom	Primary NMIBC	Intervention: WLC + BLC + WL-TURBT + BL-TURBT (HAL)	Mitomycin C/NA	12 months	68.0 (31 to 95)	95/34
				Comparator: WLC + WL-TURBT			68.0 (29 to 90)	88/32
Riedl 2001	1998 to 2000	Multicenter/Germany, Austria	Primary NMIBC	Intervention: BL-TURBT (1 g 5-ALA)	NA/PDD-TURBT after 6 weeks	60 months	67.0 (19 to 86)	36/15
				Comparator: WL-TURBT				37/14
Rolevich 2017	2008 to 2012	Single center/Republic of Belarus	Primary and recurrent NMIBC	Intervention: WLC + BLC + BL-TURBT + doxorubicin or BL-TURBT only (1 g 5-ALA)	Doxorubicin/NA	60 months	NA	134/40
				Comparator: WLC + WL-TURBT + doxorubicin or WL-TURBT only			NA	156/47
Schumacher 2010	2002 to 2005	Multicenter/Sweden	Primary and recurrent NMIBC	Intervention: WLC + BLC + WL-TURBT + BL-TURBT (1 g 5-ALA)	NA/WL-TURBT after 5 to 7 weeks in pts with pT1 G2-3 or T2	24 months	70.1 ± 10.1	103/38
				Comparator: WLC + WL-TURBT			68.9 ± 10.8	104/34
Stenzl 2010	NR	Multicenter/USA, Canada, Europe	Primary and recurrent NMIBC, only Ta/T1	Intervention: 1) WLC + mapping followed by 2nd randomization 2) BLC + mapping + BL-TURBT (0.85 g HAL)	NA/NA	9 months	68.0 ± 10.8	212/59

Table 1. Overview of included studies and synthesis (Continued)

				Comparator: WLC + WL-TURBT			69.6 ± 10.7	223/57
Stenzl 2011	NA	Multicen- ter/Germany, Austria	Primary NMIBC	Intervention: WLC + BLC + BL-TURBT (1 g 5-ALA)	NA/WL-TURBT after 2 to 4 weeks in pts with pT1 G2-3 or T2	12 months	66.0 ± 12.0	259/100
				Comparator: WLC + WL-TURBT				

^aOnly immediate postoperative instillations, no Bacille Calmette-Guerin schedule.

Abbreviations

5-ALA: 5-aminolevulinic acid; BLC: blue light cystoscopy; BL-TURBT: blue light transurethral resection of bladder tumor; G1-3: grading of tumor; HAL: hexaminolevulinic acid; NA: not available; NMIBC: non-muscle invasive bladder cancer; NR: not reported; PDD: photodynamic diagnosis; pTa, pT1: staging of tumor; pts: participants; WLC: white light cystoscopy; WL-TURBT: white light transurethral resection of bladder tumor

Table 2. Numbers of participants contributing to the interventions in the included studies

Study name	Intervention(s) and comparator(s)	Screened/ eligible (N)	Random- ized (N)	Analyzed (N): effica- cy	Analyzed (N): safety	Finishing trial (N)
Babjuk 2005	Intervention: BL-TURBT	128/122	64	60	NA	60
	Comparator: WL-TURBT		64	62	NA	62
Drăgoescu 2017	Intervention: BL-TURBT	113/113	57	57	NA	57
	Comparator: WL-TURBT		56	56	NA	56
Filbeck 2002	Intervention: BL-TURBT	301/191	151	88	NA	88
	Comparator: WL-TURBT		150	103	NA	103
Geavlete 2010	Intervention: BL-TURBT	NA/446	223	176	NA	NA
	Comparator: WL-TURBT		233	159	NA	NA
Geavlete 2012	Intervention: BL-TURBT	362/269	181	125	NA	48
	Comparator: WL-TURBT		181	114	NA	37
Gkritisios 2014	Intervention: BL-TURBT	130/104	66	48	NA	48
	Comparator: WL-TURBT		64	37	NA	37
Heer 2022	Intervention: BL-TURBT	538/533	269	209	209	NA
	Comparator: WL-TURBT		269	217	217	NA
Hermann 2011	Intervention: BL-TURBT	223/223	115	68	NA	68
	Comparator: WL-TURBT		118	77	NA	77
Karaolides 2012	Intervention: BL-TURBT	102/102	49	41	NA	41
	Comparator: WL-TURBT		53	45	NA	45
Kriegmaier 2002	Intervention: BL-TURBT	165/129	83	65	NA	65
	Comparator: WL-TURBT		82	64	NA	64
Neuzillet 2014	Intervention: BL-TURBT	151/151	72	72	NA	43
	Comparator: WL-TURBT		79	79	NA	50
O'Brien 2013	Intervention: BL-TURBT	249/185	129	86	NA	63
	Comparator: WL-TURBT		120	82	NA	67
Riedl 2001	Intervention: BL-TURBT	115/102	NA	51	51	NA
	Comparator: WL-TURBT		NA	51	51	NA

Table 2. Numbers of participants contributing to the interventions in the included studies (Continued)

Rolevich 2017	Intervention: BL-TURBT	525/377	252	174	NA	NA
	Comparator: WL-TURBT		273	203	NA	NA
Schumacher 2010	Intervention: BL-TURBT	300/279	153	141	NA	136
	Comparator: WL-TURBT		147	138	NA	134
Stenzl 2010	Intervention: BL-TURBT	814/766	382	200	421	200
	Comparator: WL-TURBT		384	202	391	202
Stenzl 2011	Intervention: BL-TURBT	381/370	192	183	187	NA
	Comparator: WL-TURBT		189	176	183	NA
Intervention total			2438	1844	868	994
Comparator total			2452	1865	842	1011
Grand total			4890	3709	1528	2005

Abbreviations

BL-TURBT: blue light transurethral resection of bladder tumor; N: number; NA: not available; WL-TURBT: white light transurethral resection of bladder tumor